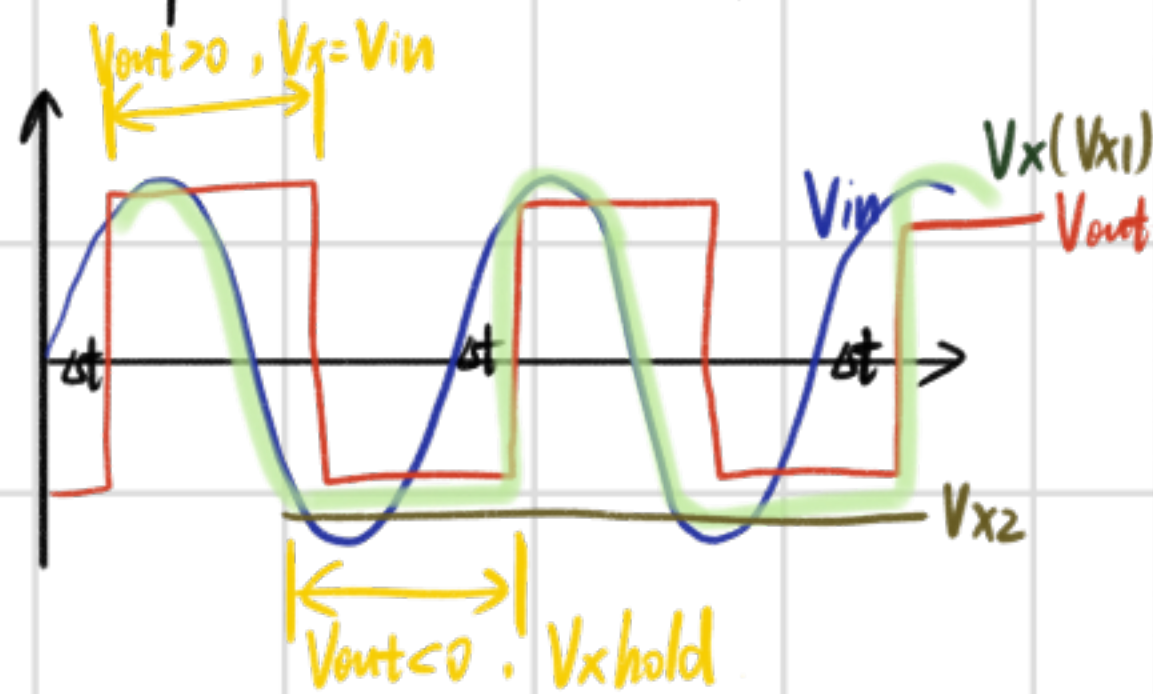


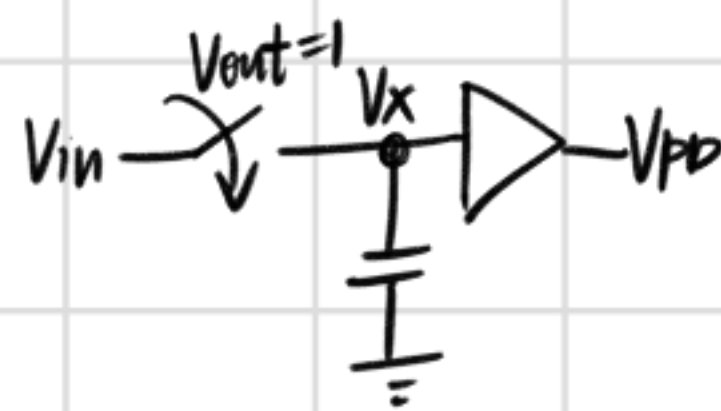
2. Sample and hold PD



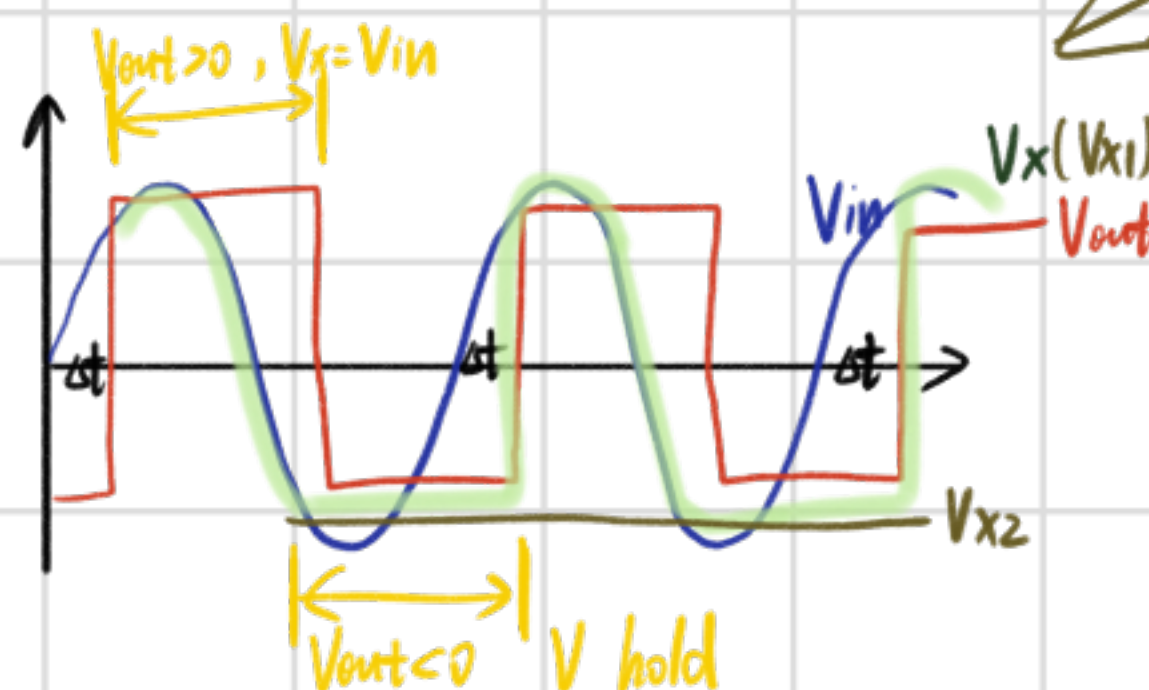
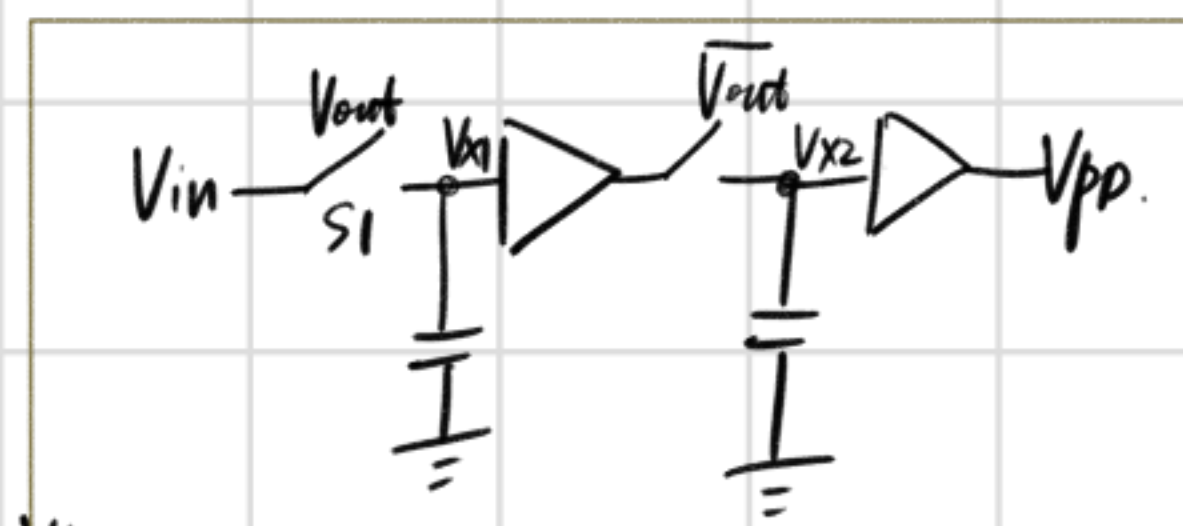
$$\Phi_{er} = 2\pi \cdot \frac{\Delta t}{T}$$

$$V_{in} = \sin(\omega_{in} t) = \sin\left(\frac{2\pi}{T} \cdot t\right)$$

V_{out} : square wave form



When V_{out} is high, switch to closed, $V_x = V_{in}$
 When V_{out} is low $\rightarrow V_x$ holds

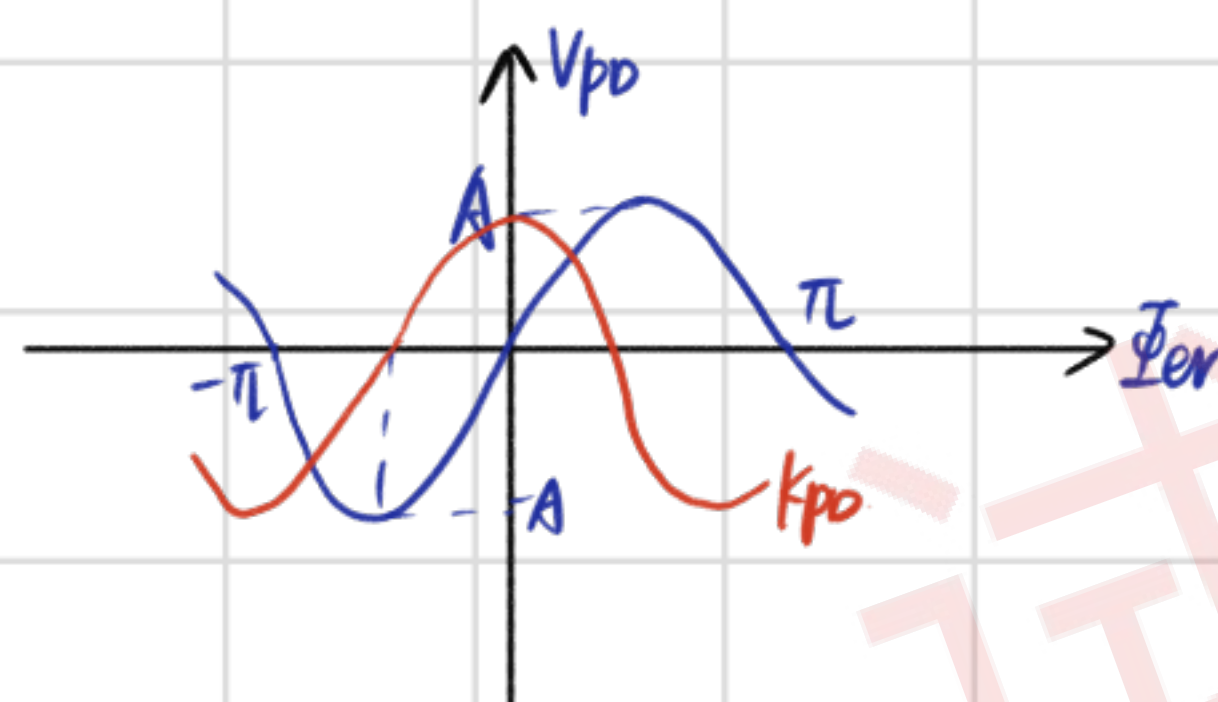


$V_{out} = \sin(\omega_{out} t)$, V_{in} is square wave

$$V_{pd} = \sin(\omega_{out} (nT + \Delta t) + \Phi_{os})$$

$$\text{or } V_{pd} = \sin(\omega_{out} (nT + \Delta t))$$

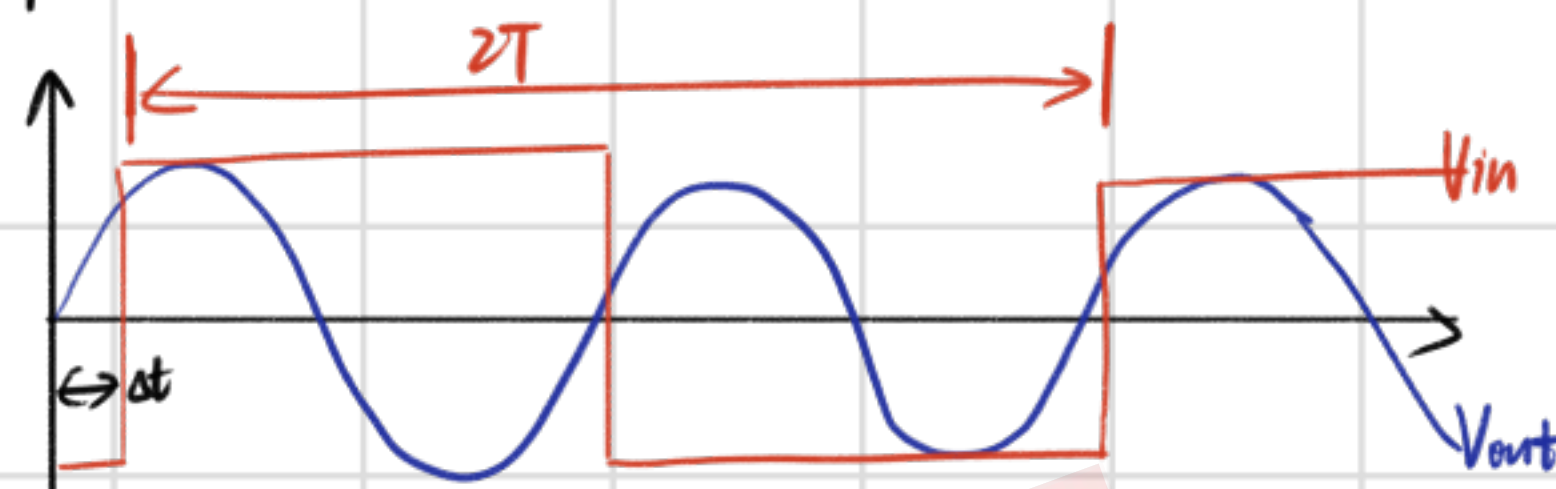
$$\text{if } \omega_{out} = \frac{2\pi}{T}, V_{pd} = A \cdot \sin\left(\frac{2\pi}{T} (nT + \Delta t)\right) = A \sin(\Phi_{er})$$



Gain of PD: $k_{pd} = A \cos(\Phi_{er})$

Linear range: $\pm \frac{\pi}{2}$

Sample & hold PD is non-linear.



$$V_{pd} = A \sin(\omega_{out} (nT + \Delta t))$$

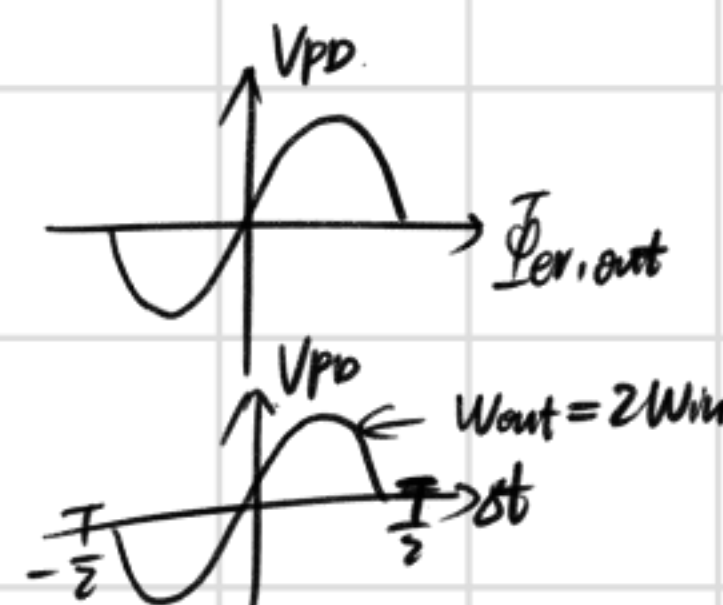
$$= A \sin\left(\frac{2\pi}{T} (2\pi T + \Delta t)\right)$$

$$= A \sin\left(2\pi \cdot \frac{\Delta t}{T}\right)$$

Φ_{er} is always defined respected to the clock period.

$$\text{respect to output: } \Delta\Phi_{out} = 2\pi \cdot \left(\frac{\Delta t}{T}\right)$$

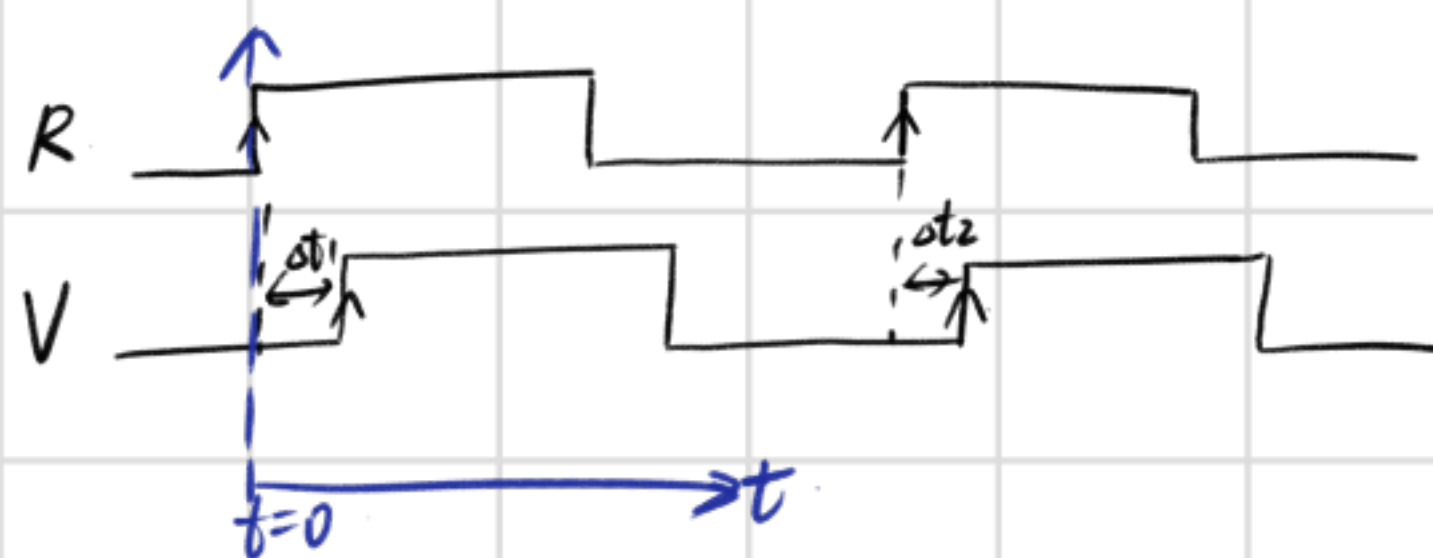
$$\text{Input: } \Delta\Phi_{in} = 2\pi \cdot \left(\frac{\Delta t}{2T}\right) = \frac{\Delta\Phi_{out}}{2}$$



"sub-sample" PD

Digital Phase Error Detector

1. Exclusive-OR (EXOR) based PD

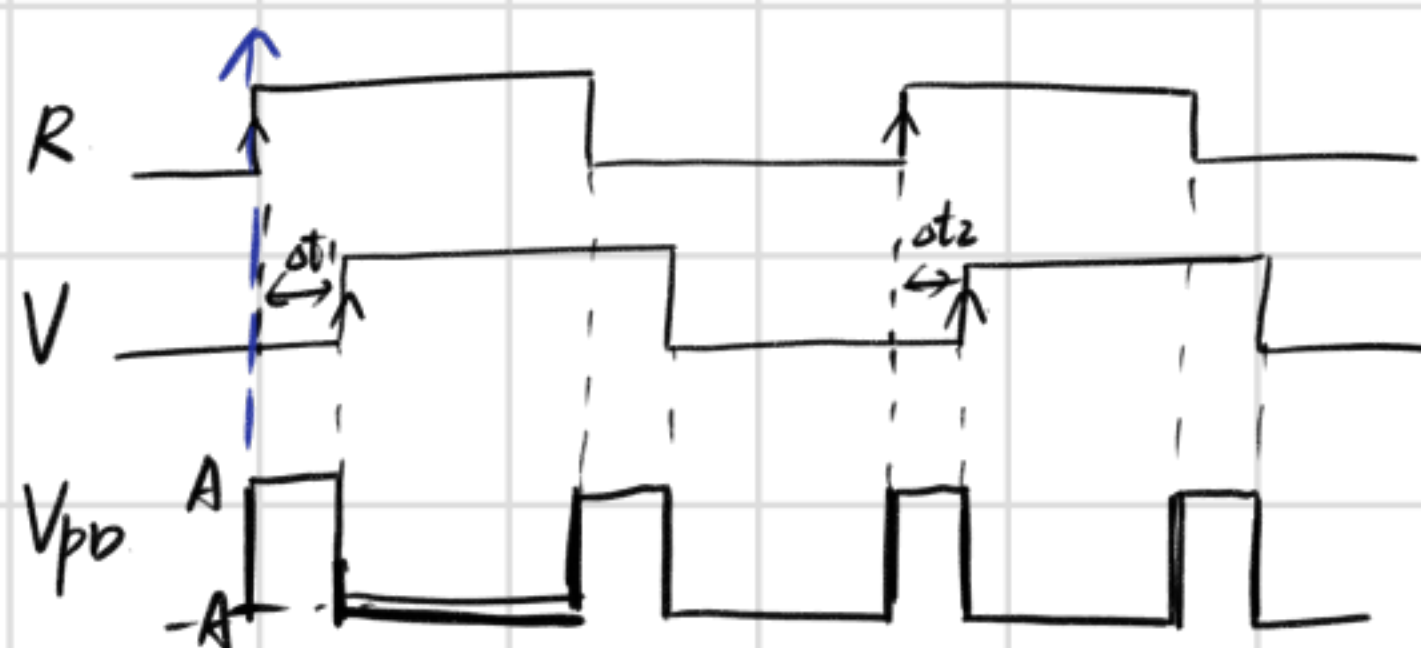


Phase Error is measured with a reference clock.

Here, the reference clock is R with period T.

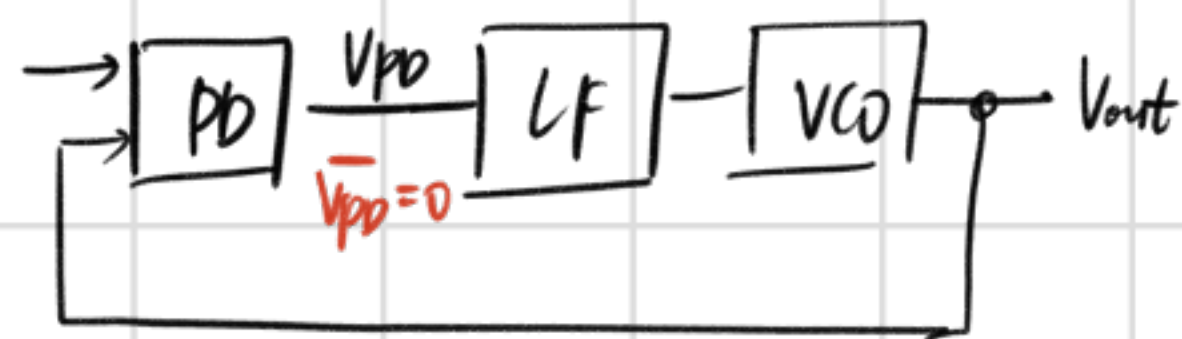
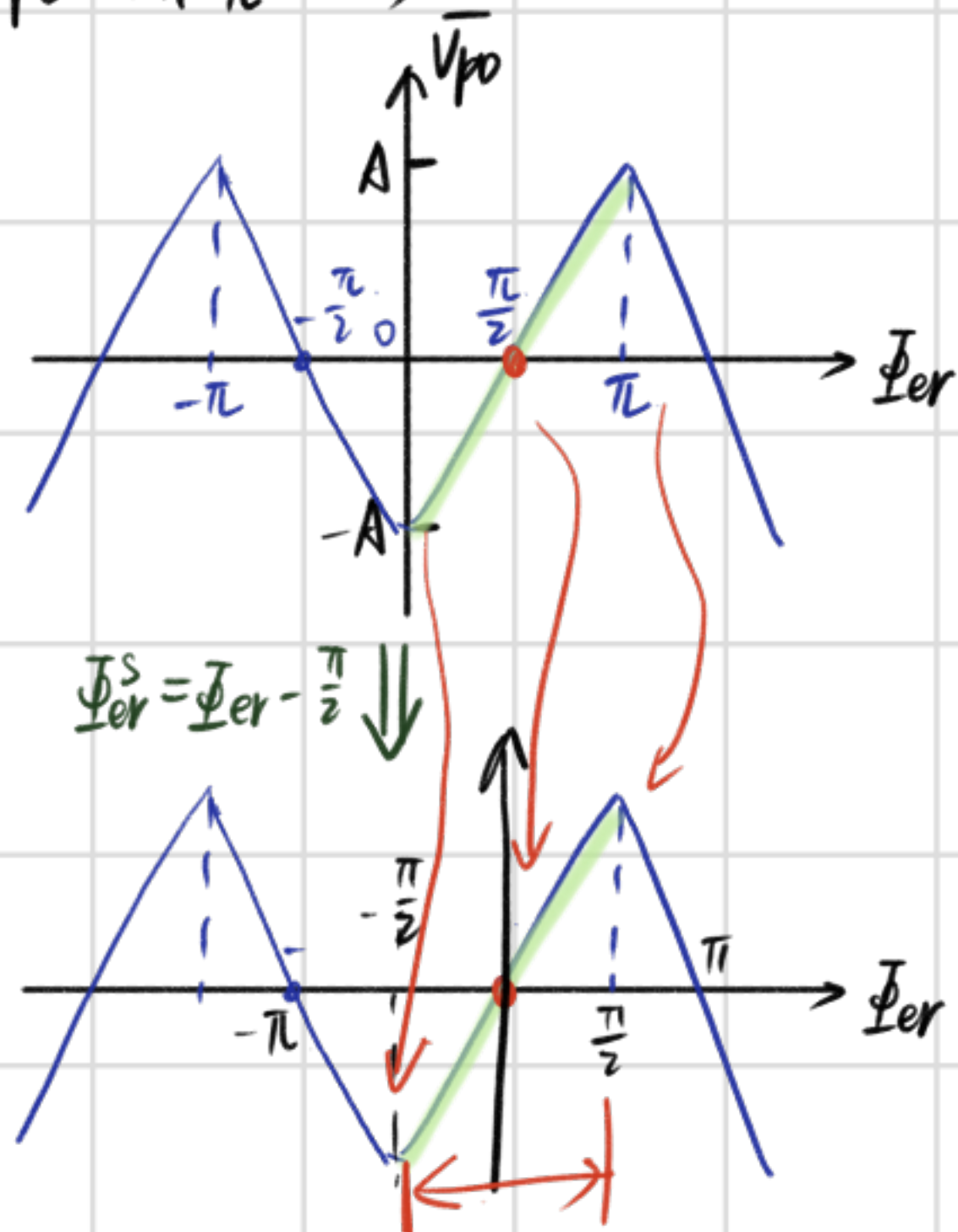
$$f_{ref} = \frac{1}{T}$$

$$\Phi_{er} = 2\pi \cdot \frac{\Delta t}{T}$$



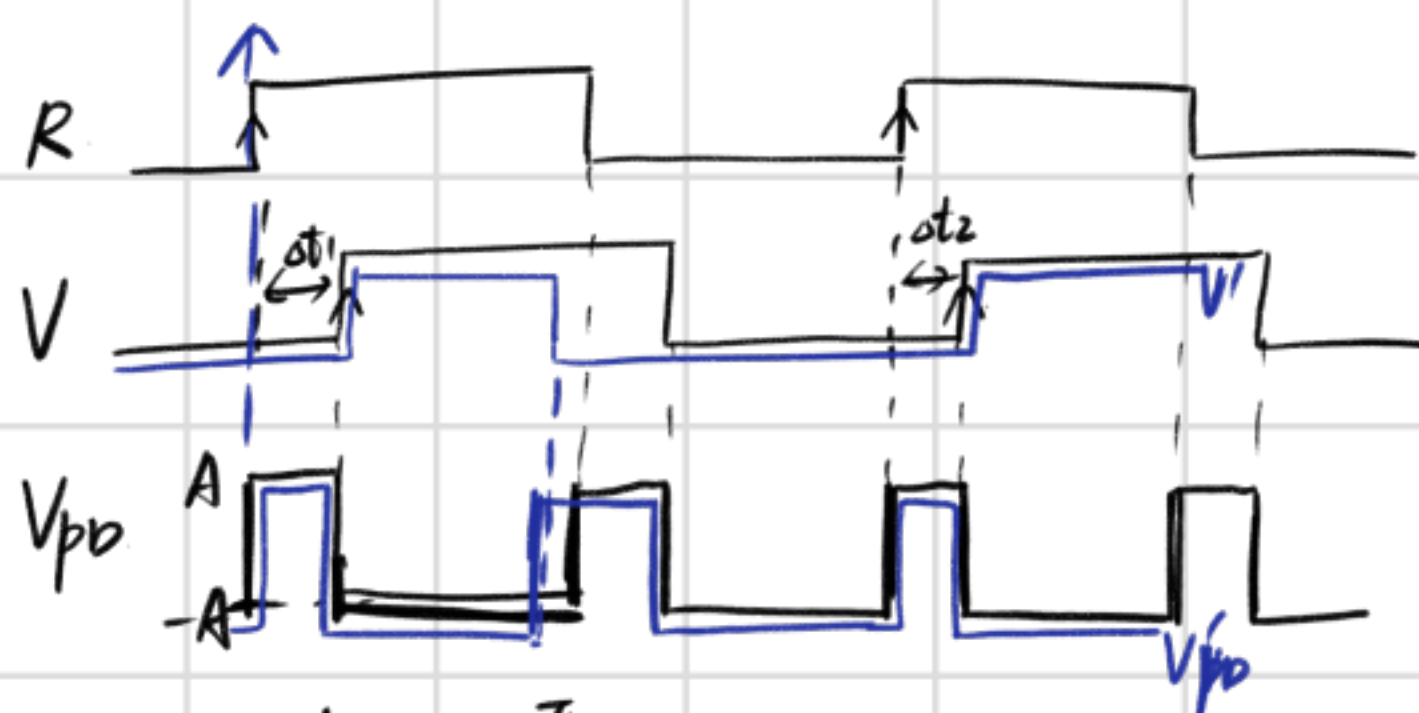
$$\overline{V_{pd}} = \frac{A \cdot \Delta t - A \left(\frac{T}{2} - \Delta t\right)}{T/2} = A \left(4 \frac{\Delta t}{T} - 1\right) = A \left(\frac{2\Phi_{er}}{\pi} - 1\right)$$

$$\bar{V}_{pd} = A \left(\frac{2\bar{\Phi}_{er}}{\pi} - 1 \right)$$



Linear Range: $\pm \frac{\pi}{2}$

Gain of PD, $K_{pd} = \frac{2A}{\pi} = \frac{2}{\pi} A \text{ (V} \cdot \text{rad}^{-1})$

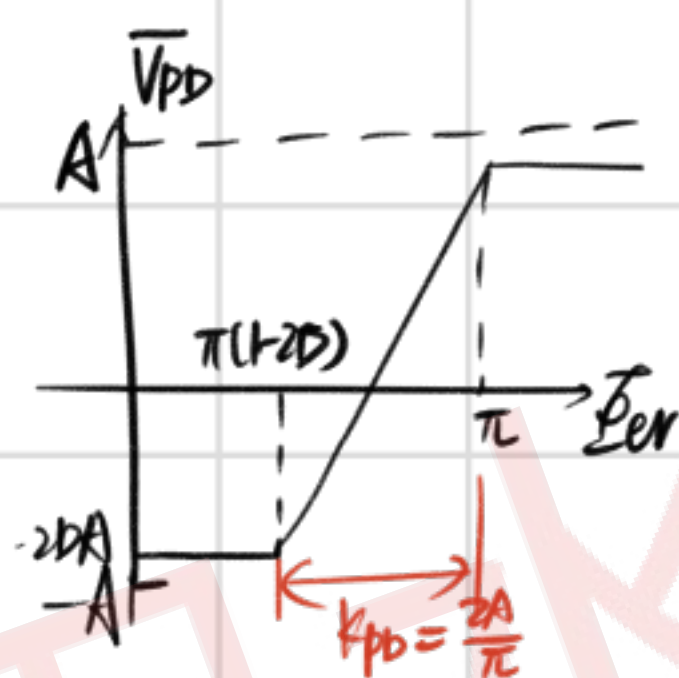
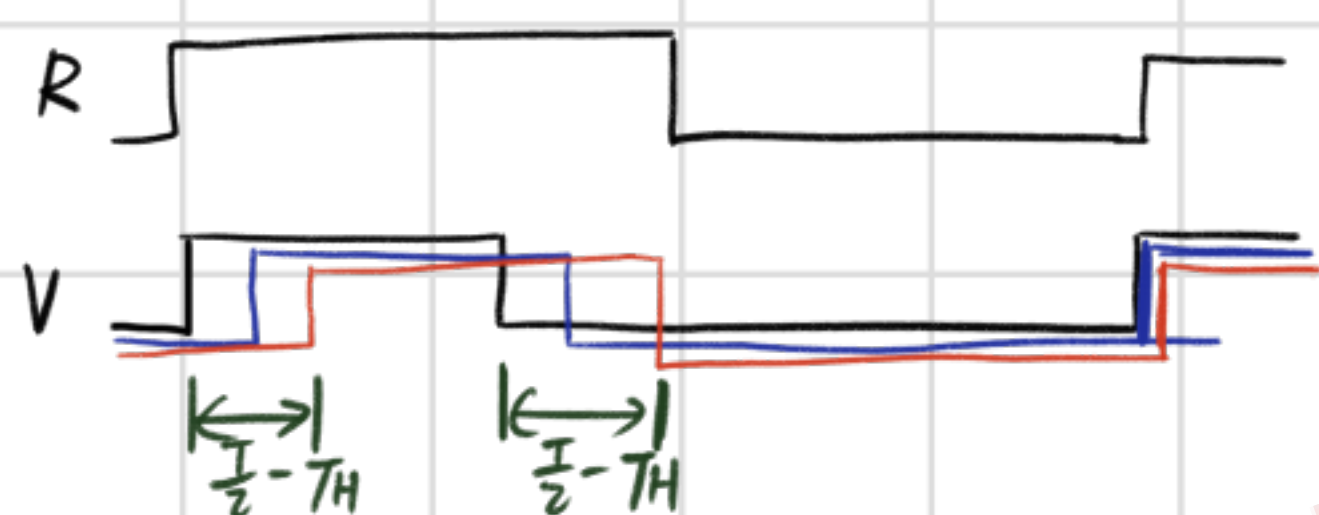


Depend on Duty Cycle!

$$\text{Duty cycle: } D = \frac{T_H}{T}$$

In process of phase error, $\bar{\Phi}_{er}' = 2\pi \cdot \frac{\delta t_1}{T}$ and $\bar{\Phi}_{er}'' = 2\pi \cdot \frac{\delta t_2}{T}$

$$\bar{\Phi}_{er} = 2\pi \cdot \frac{\delta t}{T}, \quad \bar{V}_{pd} = \bar{V}_{pd}'$$



Linear Range: $[\pi(1-D), \pi]$

for $0 \leq \delta t \leq \frac{T}{2} - DT$

$$\bar{V}_{pd} = -2DA$$

for $(\frac{T}{2} - DT) \leq \delta t \leq \frac{T}{2}$

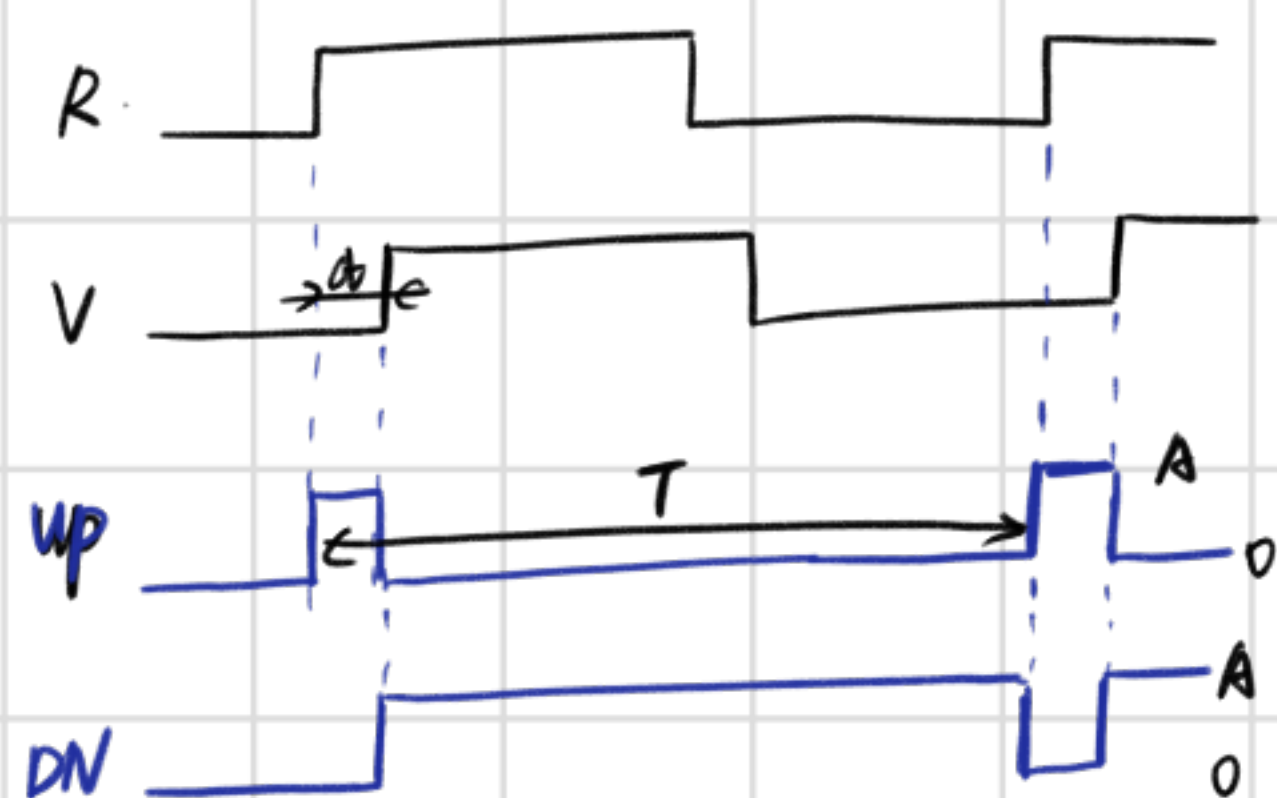
$$\bar{V}_{pd} = -2A \left[1 - D - 2 \cdot \frac{\delta t}{T} \right]$$

PLL will lock for $\bar{V}_{pd} = 0$

$$1 - D - 2 \cdot \frac{\delta t}{T} = 0, \quad \delta t = \frac{T}{2} (1 - D)$$

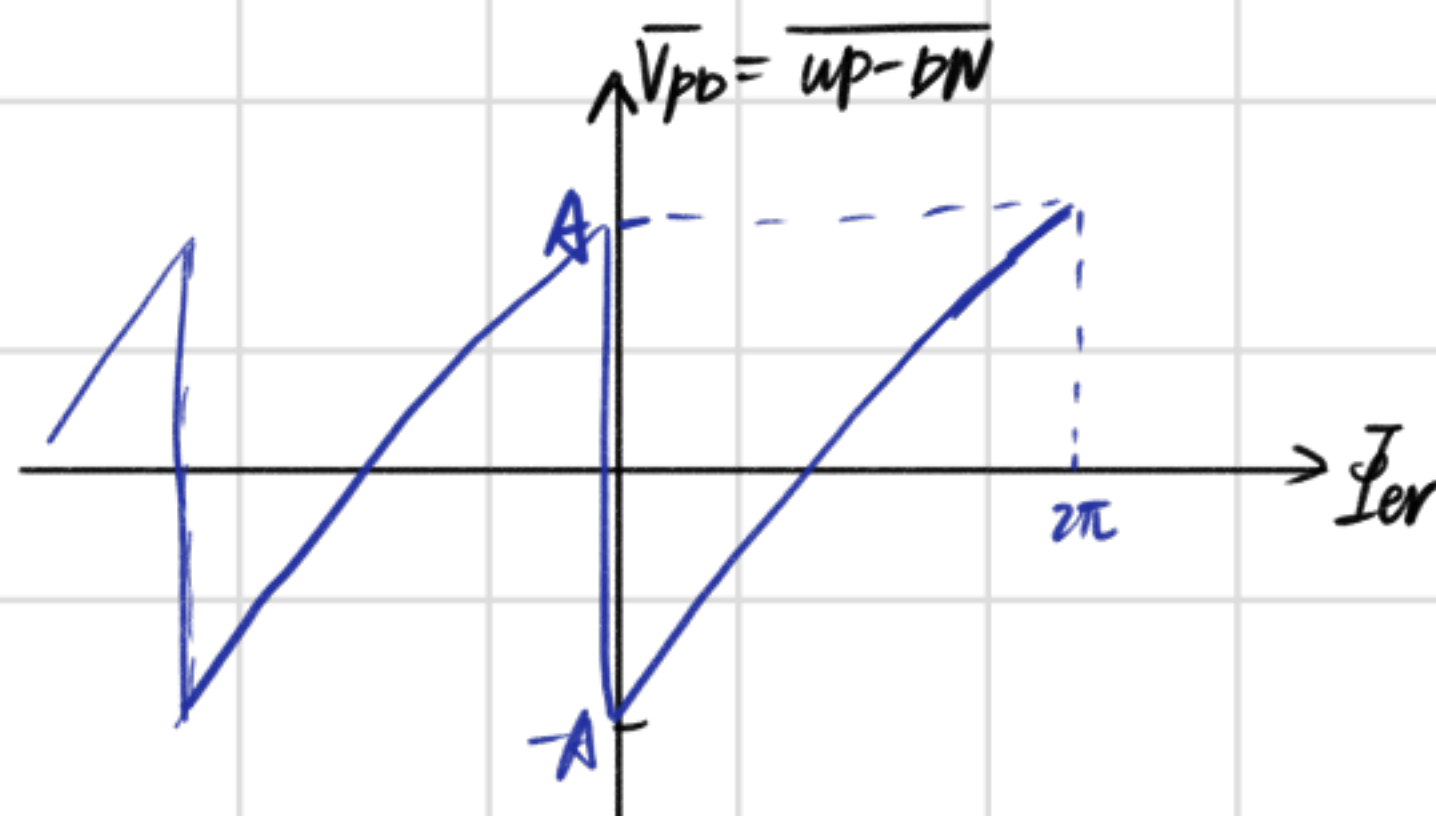
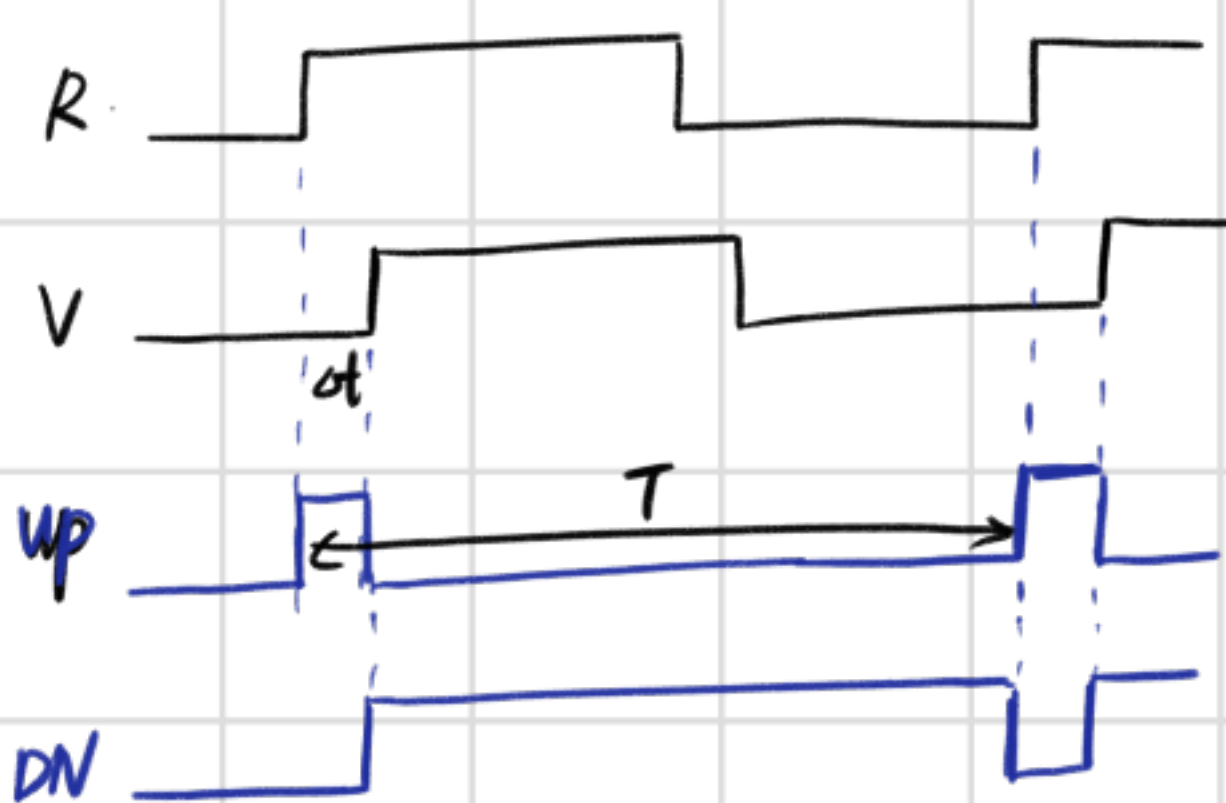
$$\text{Earlier: } D = \frac{1}{2}, \quad \delta t = \frac{T}{2} (1 - D) = \frac{T}{4}$$

2. S-R flip-flop based PD



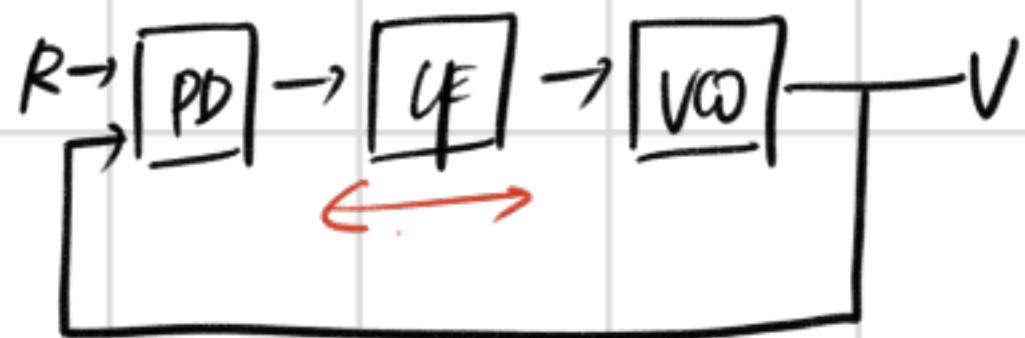
$$\bar{V}_{pd} = \overline{up - dn} = \frac{A \cdot \delta t - A(T - \delta t)}{T} = A \left(2 \frac{\delta t}{T} - 1 \right)$$

$$= A \left(\frac{2\bar{\Phi}_{er}}{2\pi} - 1 \right) = A \cdot \left(\frac{\bar{\Phi}_{er}}{\pi} - 1 \right)$$

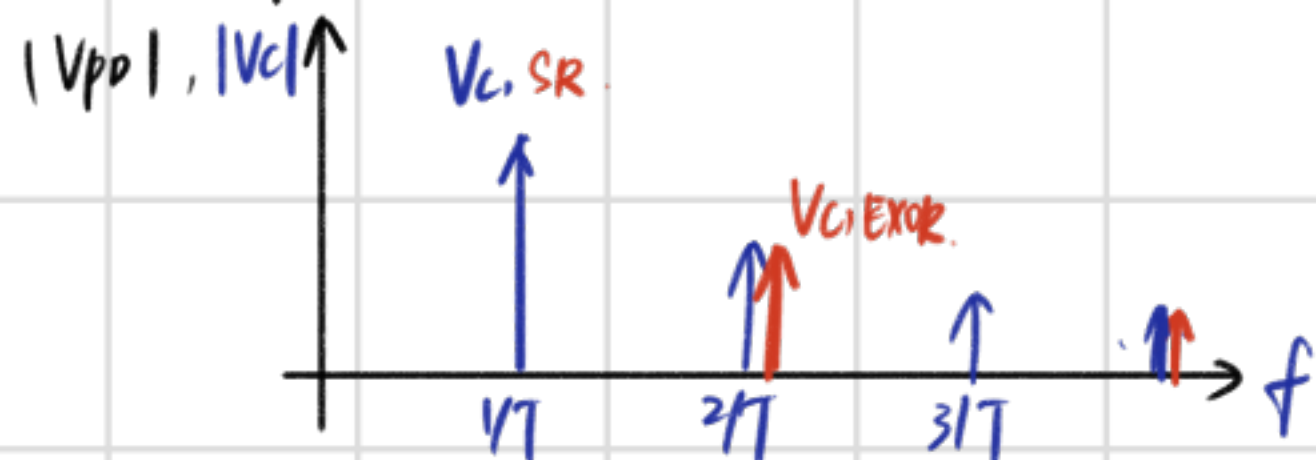


Linear Range: $[0, 2\pi]$

$$\text{Gain } K_{pd} = \frac{2A}{2\pi} = \frac{A}{\pi} \text{ V} \cdot \text{rad}^{-1}$$



V_c changes twice faster in EXOR-PD as compared to S-R ff based PD.



- Loop filters $V_{pd, EXOR}$ more efficiently than $V_{pd, SR}$.

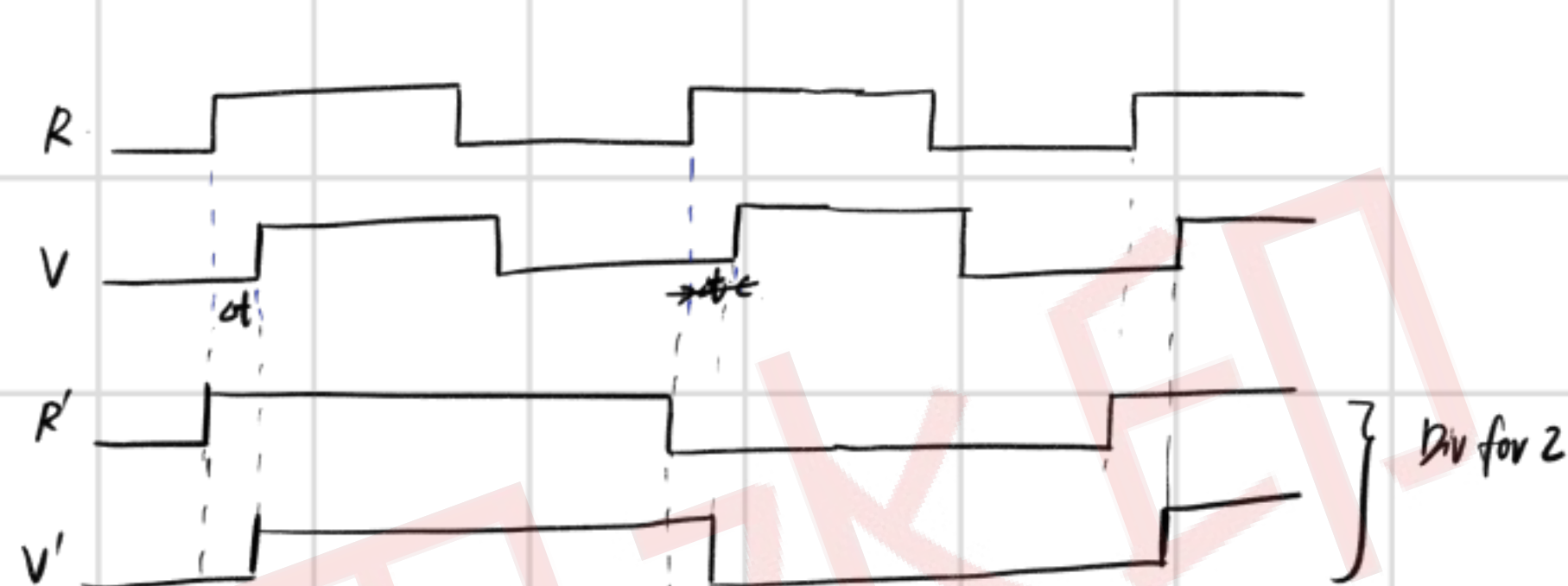
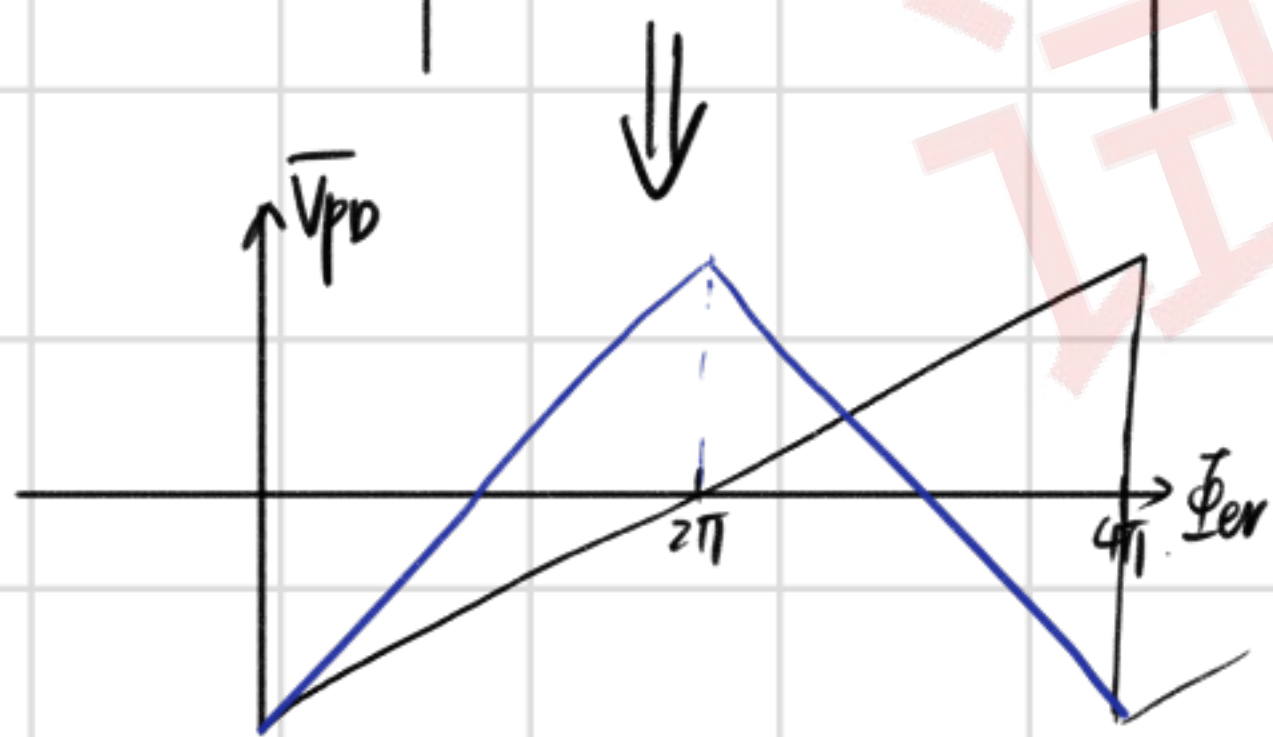
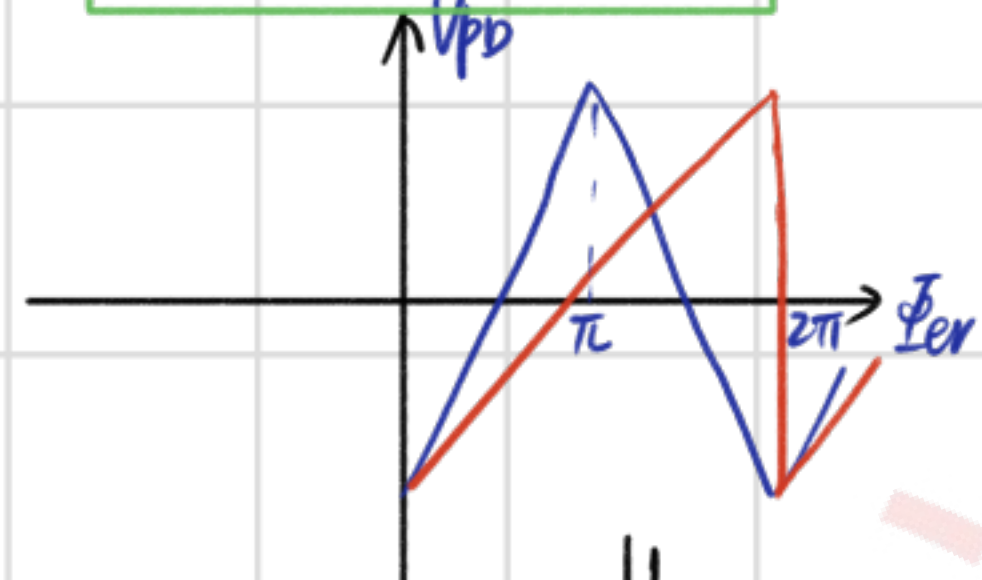
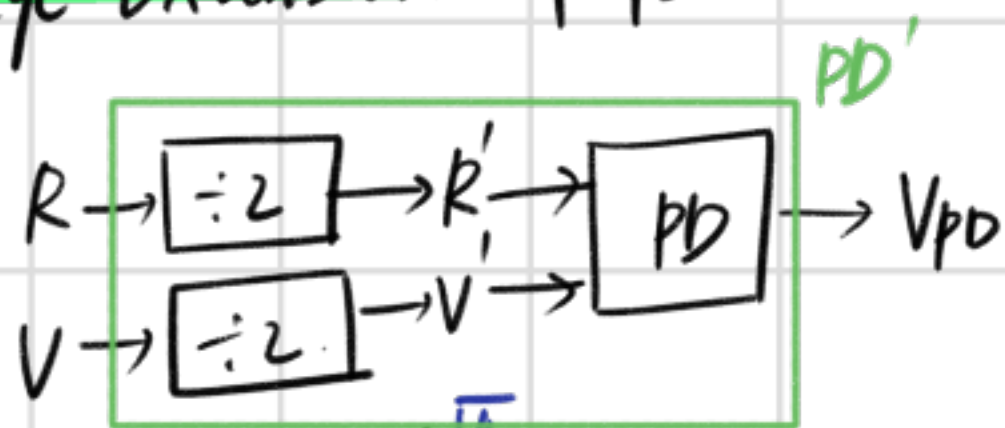
- V_c modulates output frequency.

$$f_o \rightarrow (f_o \pm \frac{1}{T}) \text{ (SR)} \quad (f_o \pm \frac{2}{T}) \text{ (EXOR)}$$

$$(f_o \pm f_{ref}) \quad (f_o \pm 2f_{ref})$$

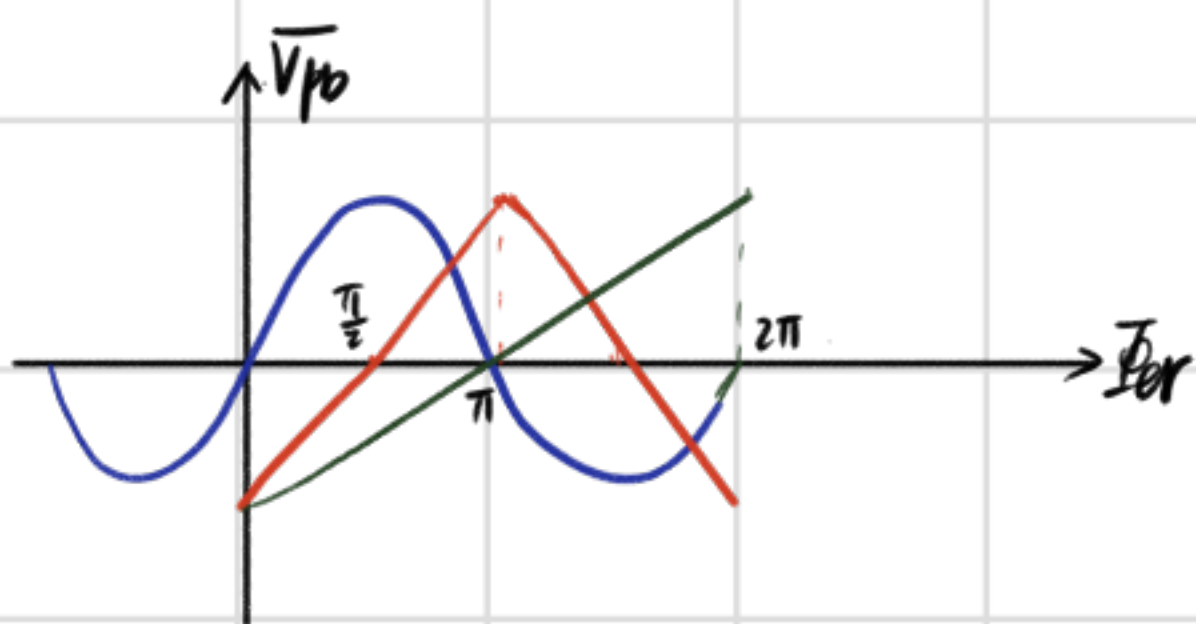
magnitude of $(f_o \pm 2f_{ref})$ is lesser than magnitude of $(f_o \pm f_{ref})$ at VCO's output.

Range Extension of PD



2π phase error for R', V' signals is same as 4π phase error for R, V signals

Phase Frequency Detector



$f_R \neq f_V$:

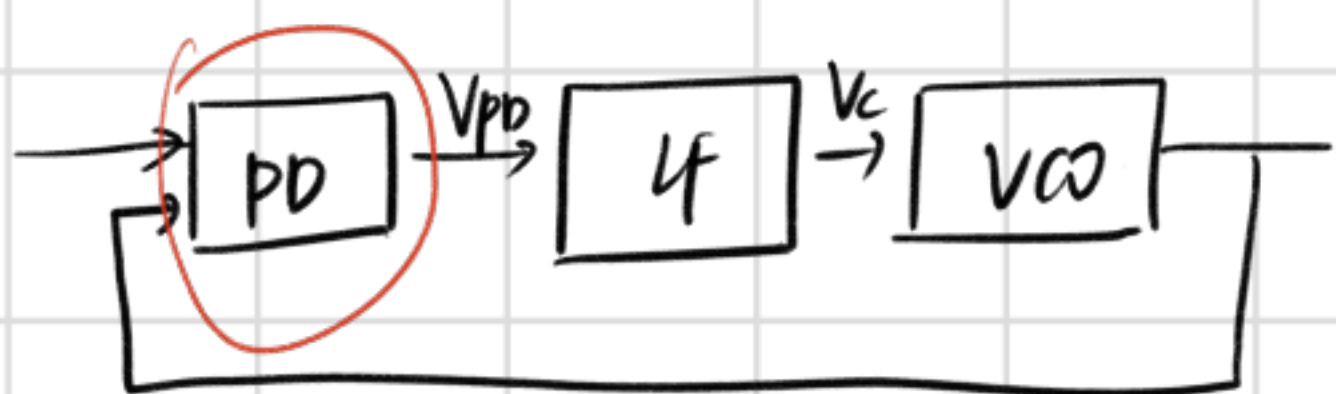
$$\Phi_{er}(t) = \int 2\pi (f_R - f_V) dt \quad \uparrow$$

R: 略

V: 略

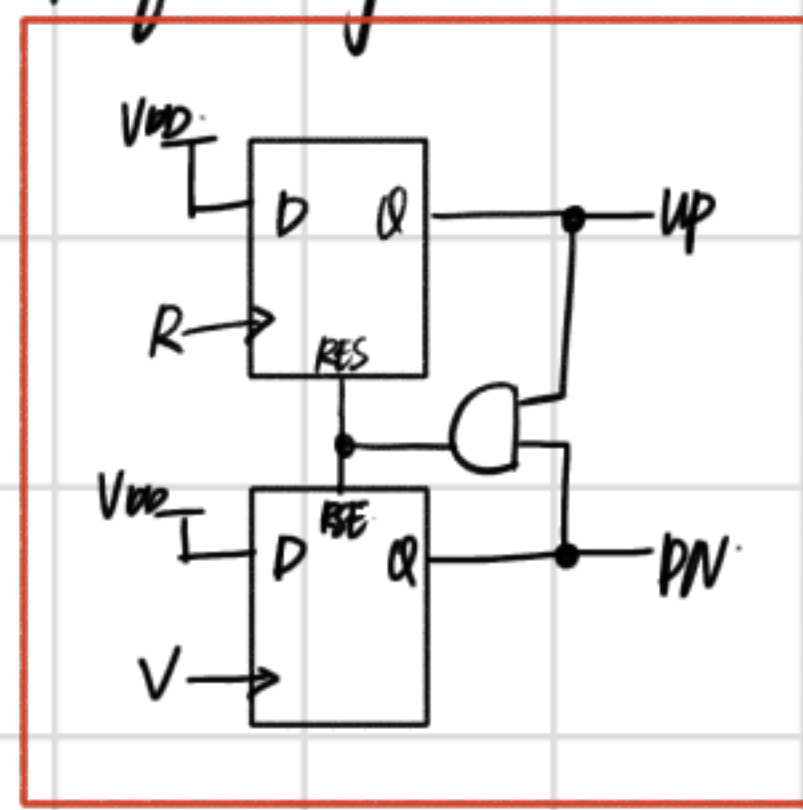
$$\frac{f_R}{f_V} = \frac{m}{n} \Rightarrow nT_V = mT_R$$

不会遇到大的输入 of problem !!



Need a block that give the correct frequency error value then to detector the frequency error.

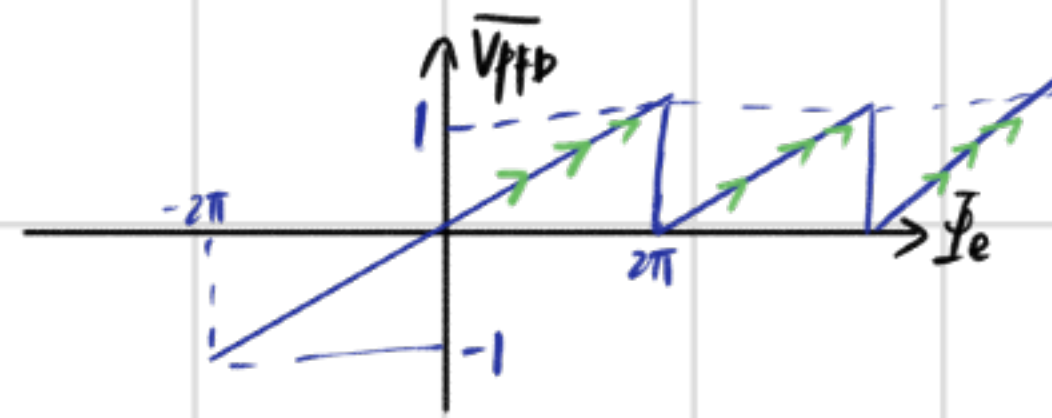
Phase frequency Detector (PFD)



$$\overline{V_{PFD}} = \overline{UP - DN} = A \cdot \frac{\Delta t}{T} = \frac{A}{2\pi} \Delta \phi_e$$

if $A=1$, $\overline{V_{PFD}} = \frac{1}{2\pi} \Delta \phi_e$

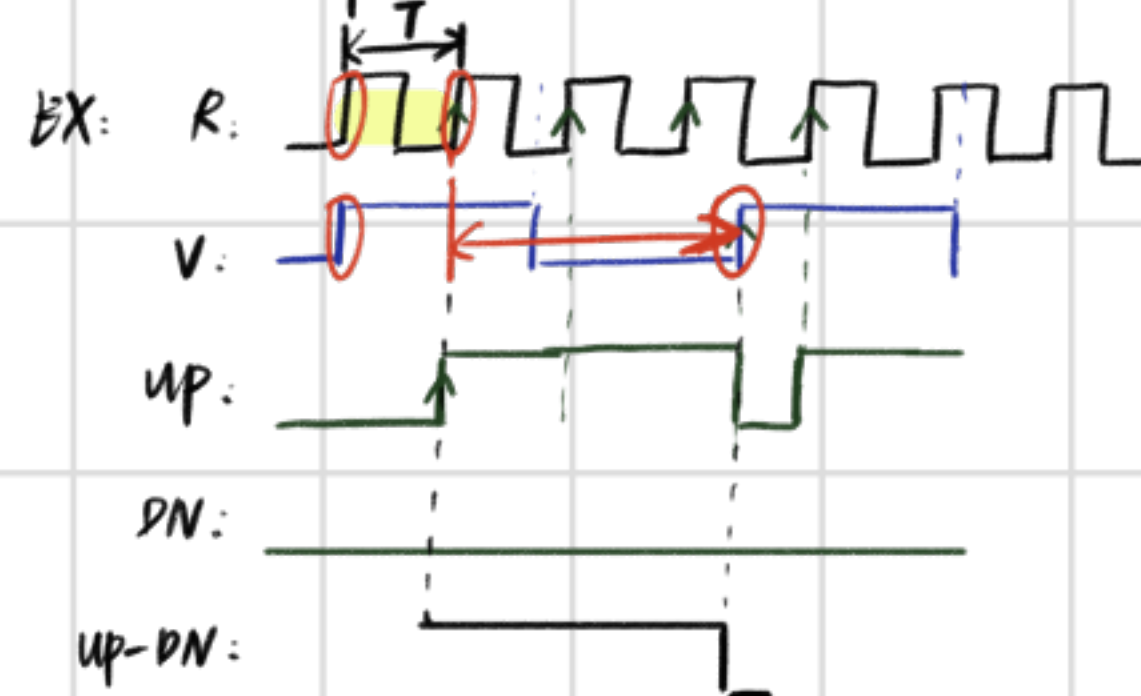
tackle this problem



frequency error > 0 , $\overline{V_{PFD}} > 0$
frequency error < 0 , $\overline{V_{PFD}} < 0$

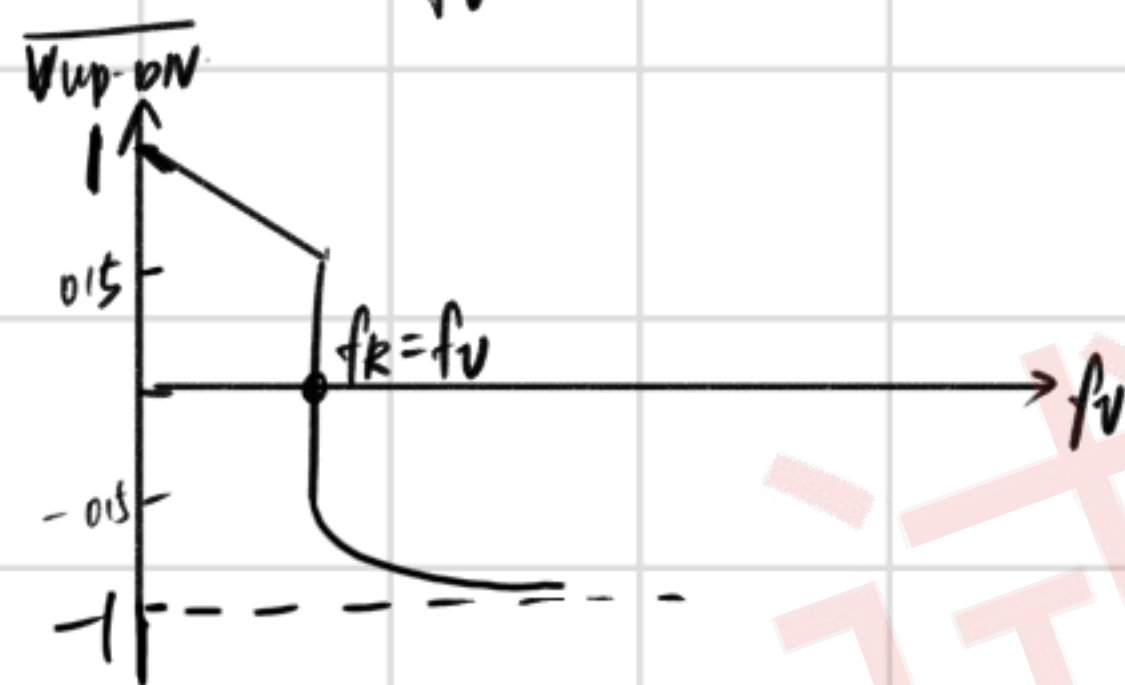
Linear Range: $[-2\pi, 2\pi]$

Gain: $K_{PD} = \frac{1}{2\pi} V \cdot \text{rad}^{-1}$

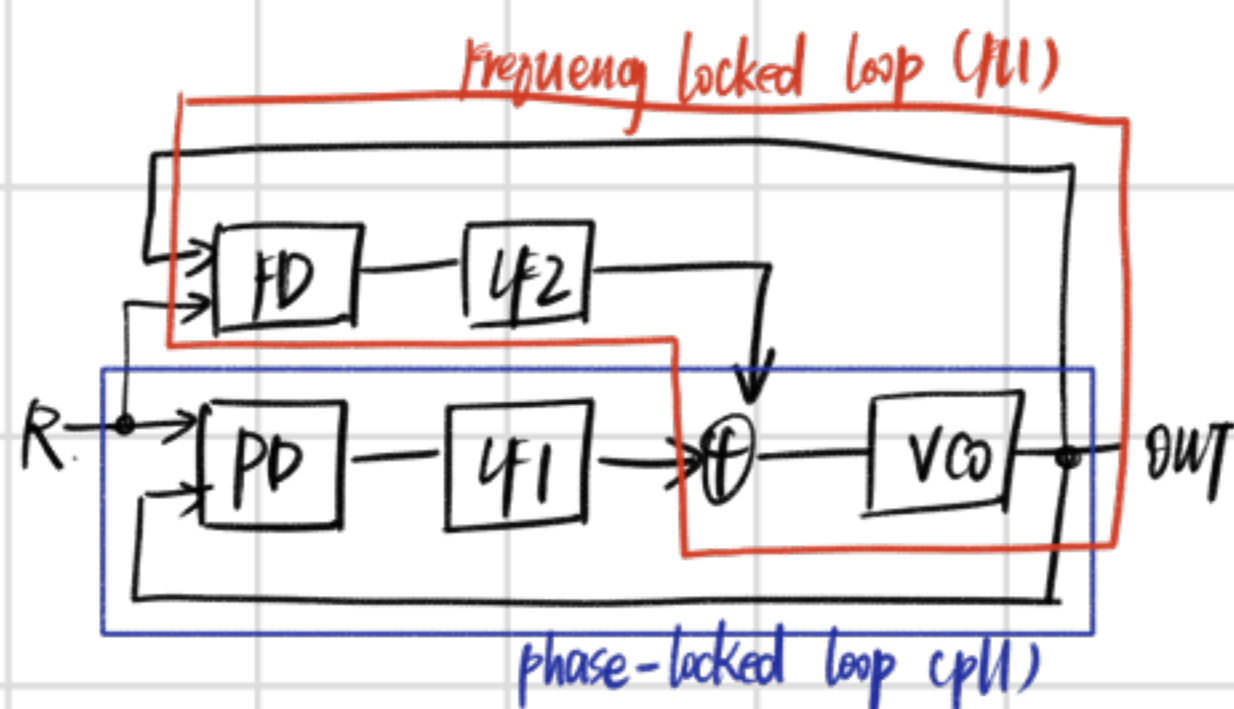


Frequency Transfer Characteristic

$$\overline{V_{UP-DN}} = \begin{cases} 1 - \frac{0.5 f_v}{f_R}, & \text{if } f_R > f_v \\ \frac{0.5 f_R}{f_v} - 1, & \text{if } f_v > f_R \end{cases}$$



Digital Phase Detector

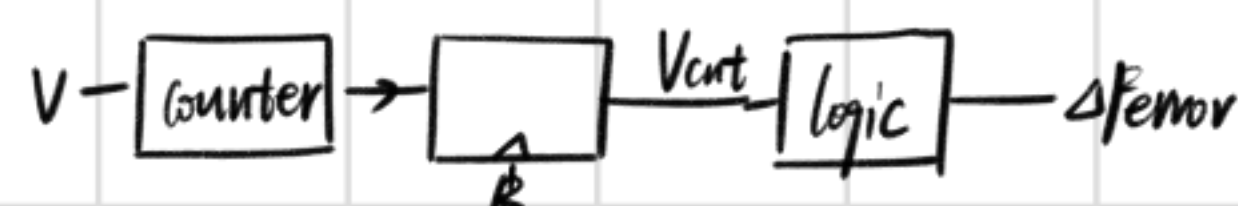


there, FD means frequency detector

- VCO is controlled by both PLL and FLL $\Rightarrow$ problem.
so: PLL has very low bandwidth compared to PLL.

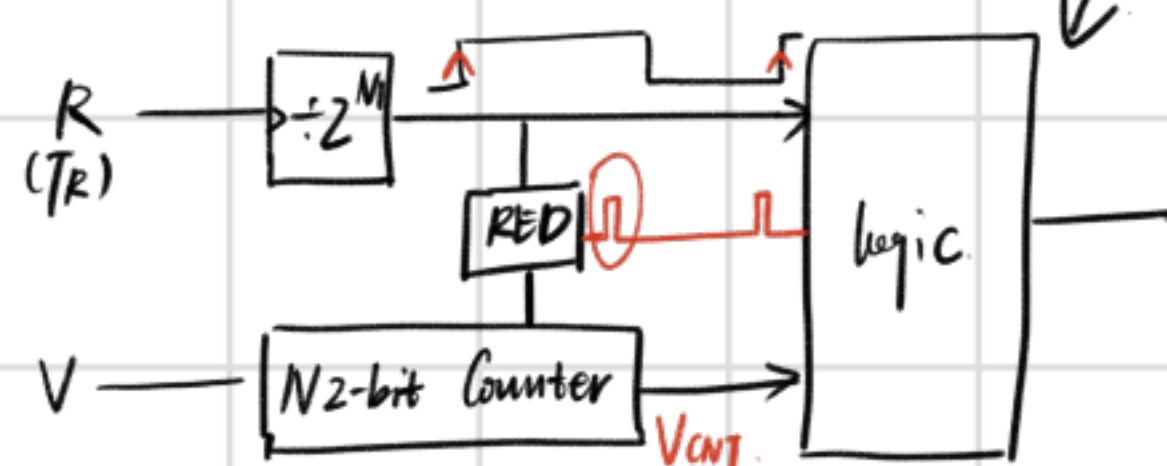
Frequency detector: $\begin{matrix} R \\ V \end{matrix} \rightarrow \boxed{FD}$

- counter based FD



In $1 T_R$, count value = 10 $\Rightarrow 1/T_R = 10 T_V$ (example)

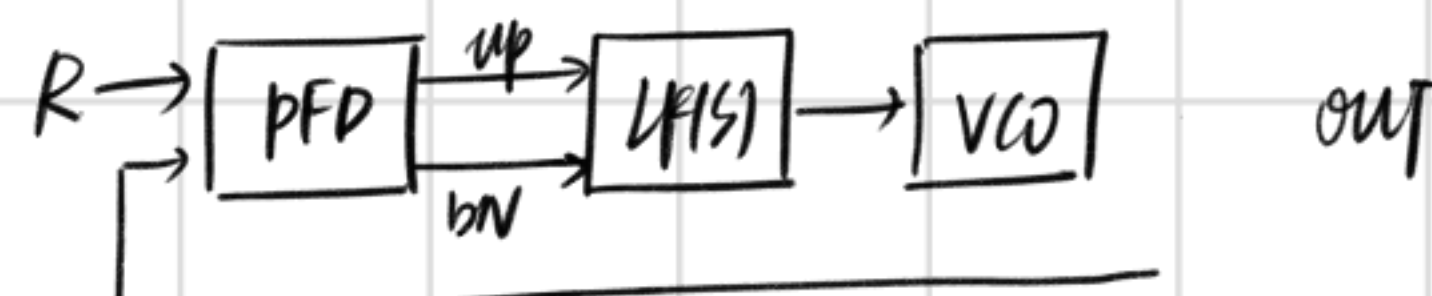
Estimated output frequency $f_{out} = \frac{1}{T_V} = \frac{f_0}{T_R} = 10 f_R$
在一个周期内, 存在上采样误差



$$2^M T_R = \{ V_{CNT}(n(2^M T_R)) - V_{CNT}((n-1)(2^M T_R)) \} \times T_V$$

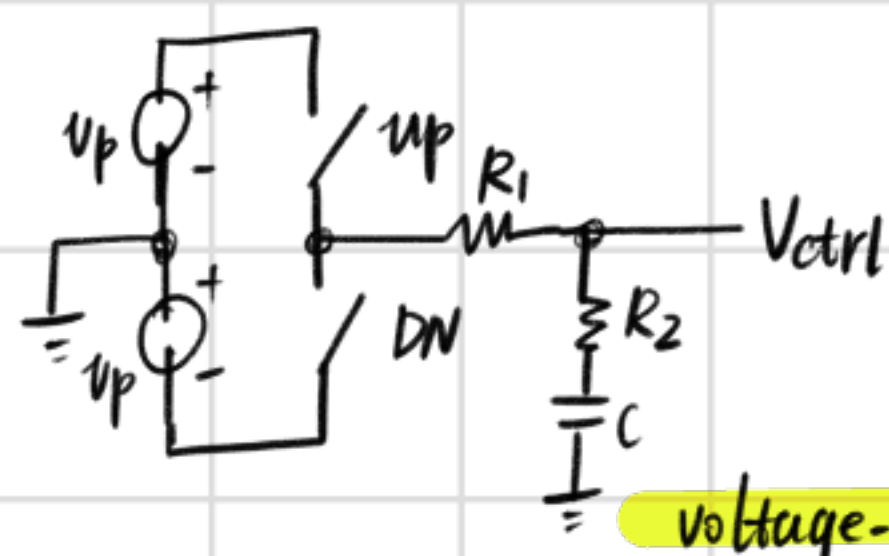
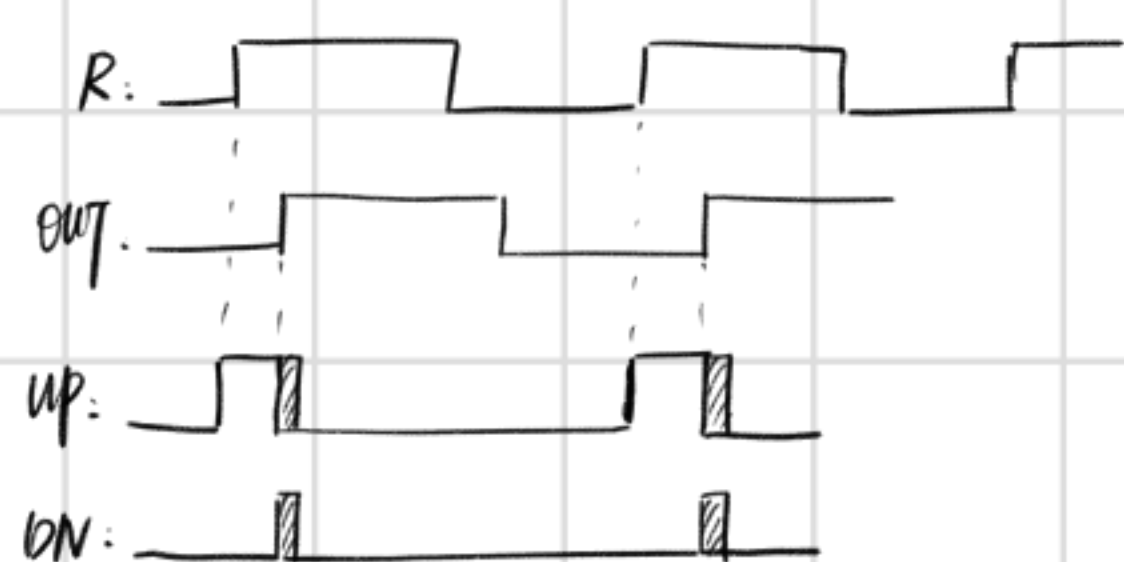
$$f_v = \frac{1}{T_V} = \frac{\Delta V_{CNT}}{2^M \cdot T_R}$$

Basic PLL block diagram.

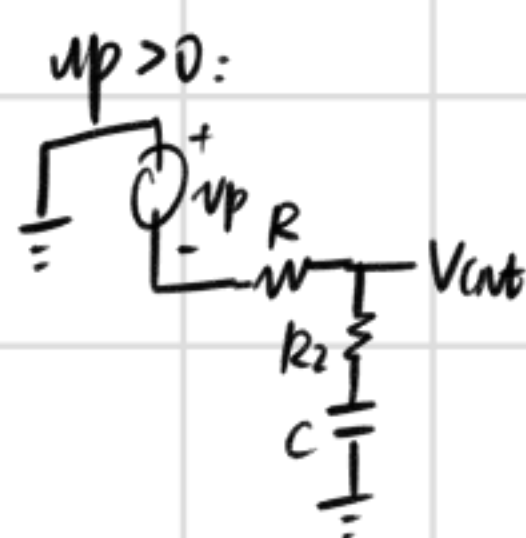


How convert up & DN signal to voltage signals?

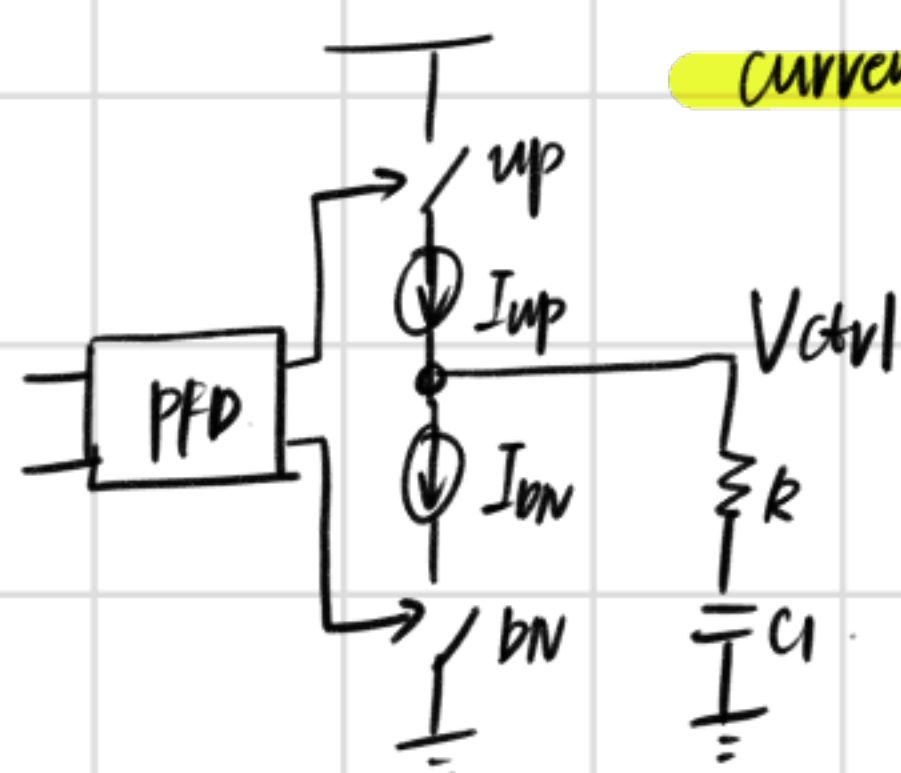
Charge Pump PLL



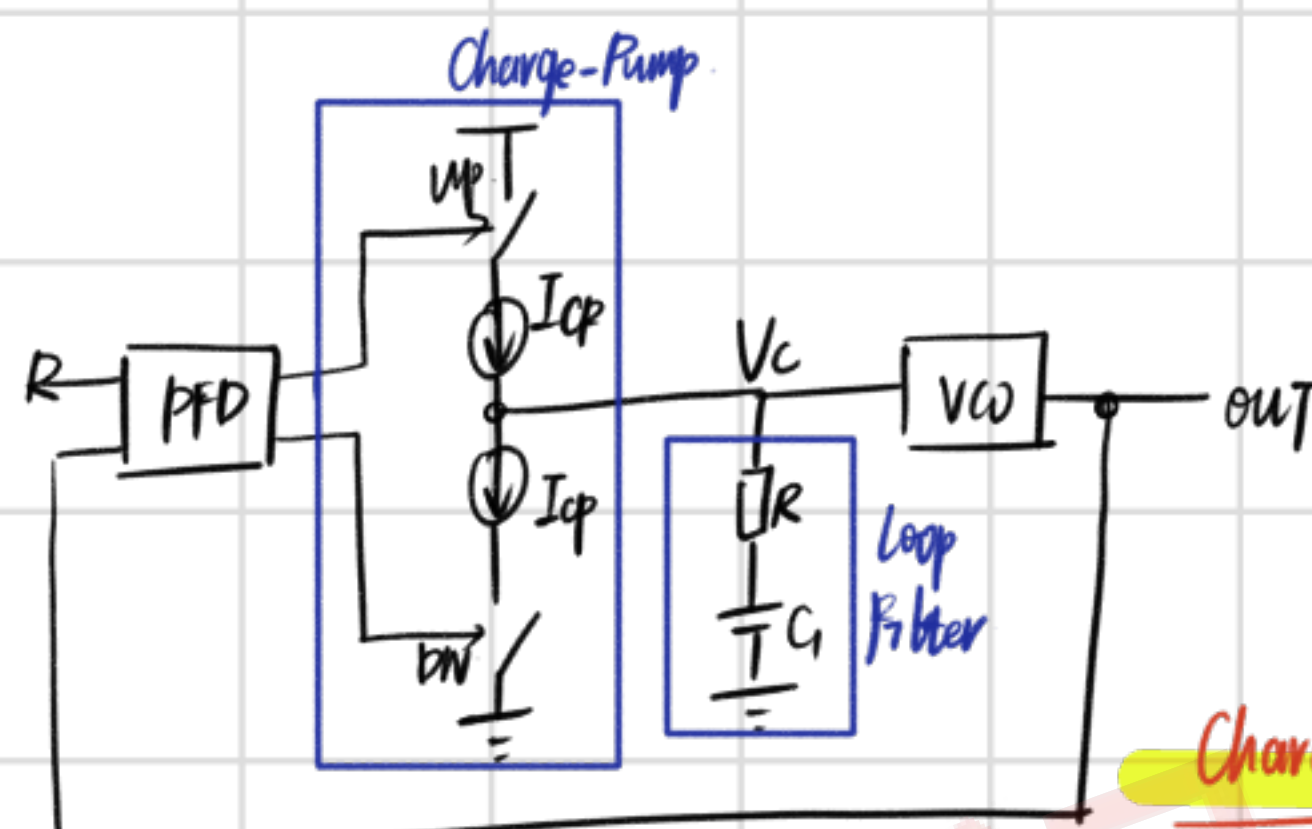
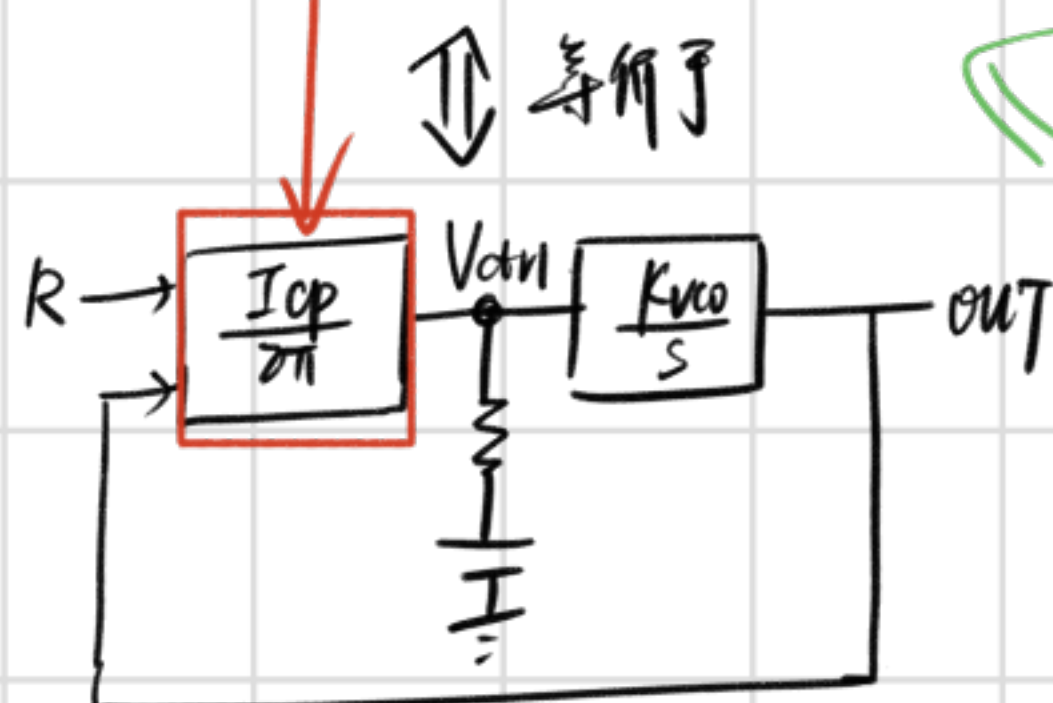
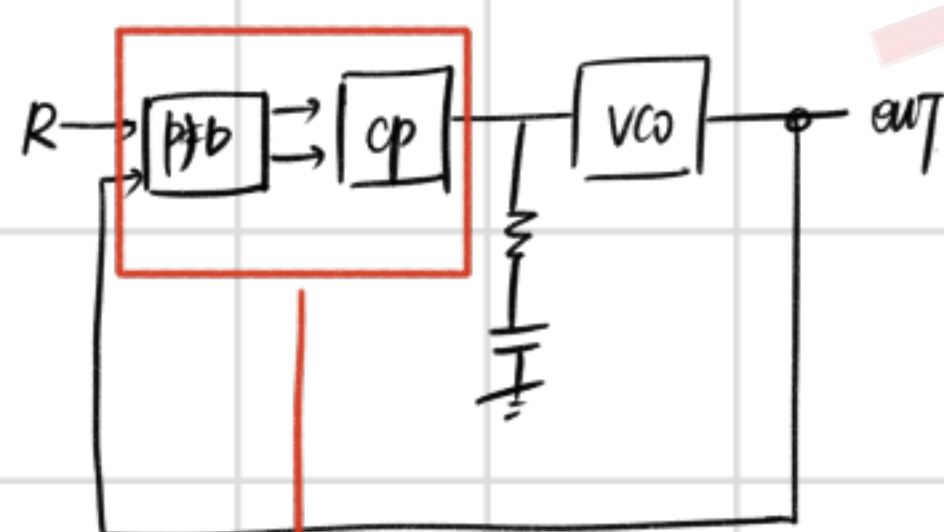
voltage-source based



current-source based



$$V_{ctrl}(t) = I_{cp}R + \frac{1}{C} \int I_{cp} \cdot dt$$



Charge Pump based PLL

$$V_{ctrl}(nT_R) = V_{ctrl}((n-1)T_R) + \frac{I_{cp} \cdot \Delta t(n-1)T_R}{C}$$

$$= V_{ctrl}((n-1)T_R) + \frac{I_{cp} \cdot T_R}{2\pi C} \cdot (2\pi \frac{\Delta t}{T_R})$$

$$V_{ctrl}(nT_R) = V_{ctrl}((n-1)T_R) + \frac{I_{cp} \cdot T_R}{2\pi C} \cdot \Phi_{err}((n-1)T_R)$$

$$\rightarrow V_{ctrl}(z) = z^{-1} V_{ctrl}(z) + \frac{I_{cp} \cdot T_R}{2\pi C} \cdot z^{-1} \cdot \Phi_{err}(z)$$

$$\frac{V_{ctrl}(z)}{\Phi_{err}(z)} = \frac{I_{cp} \cdot T_R}{2\pi C} \cdot \frac{z^{-1}}{1-z^{-1}} = \frac{I_{cp} \cdot T_R}{2\pi C} \cdot \frac{1}{z-1}$$

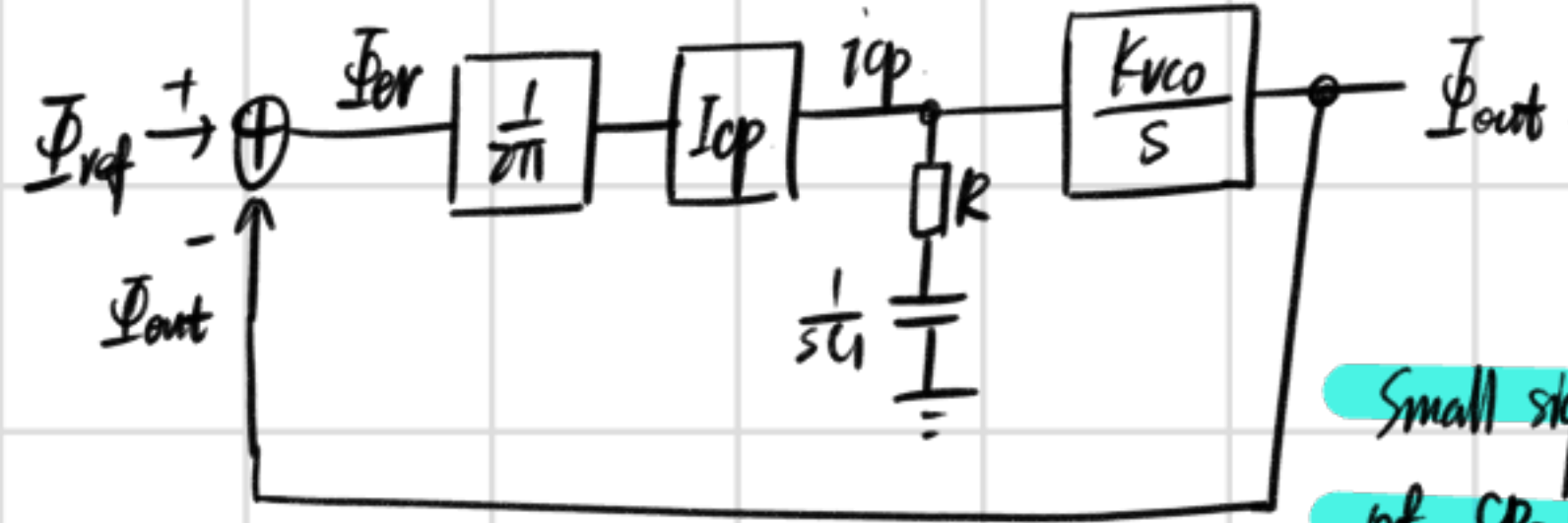
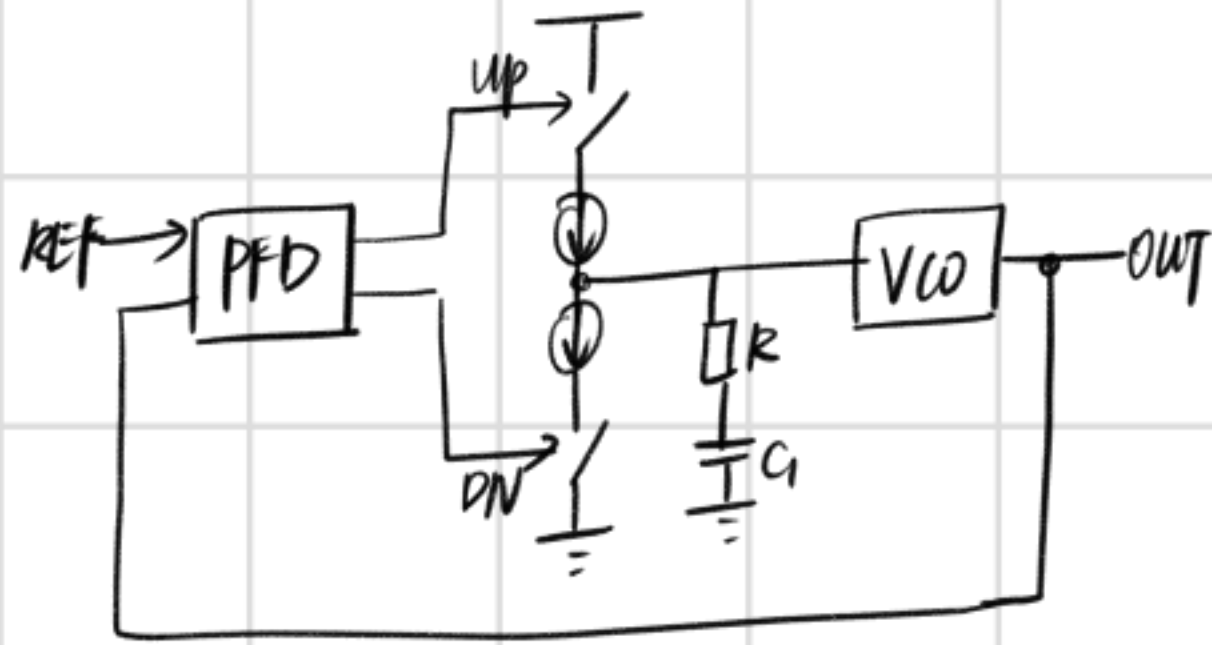
Transfer Function

$$\text{for } |W T_R| \ll 1, e^{sT_R} \approx 1 + sT_R$$

$$\frac{V_{ctrl}(z)}{\Phi_{err}(z)} = \frac{I_{cp} \cdot T_R}{2\pi C} \cdot \frac{1}{1 + sT_R - 1} = \frac{I_{cp}}{2\pi C} \cdot \frac{1}{s}$$

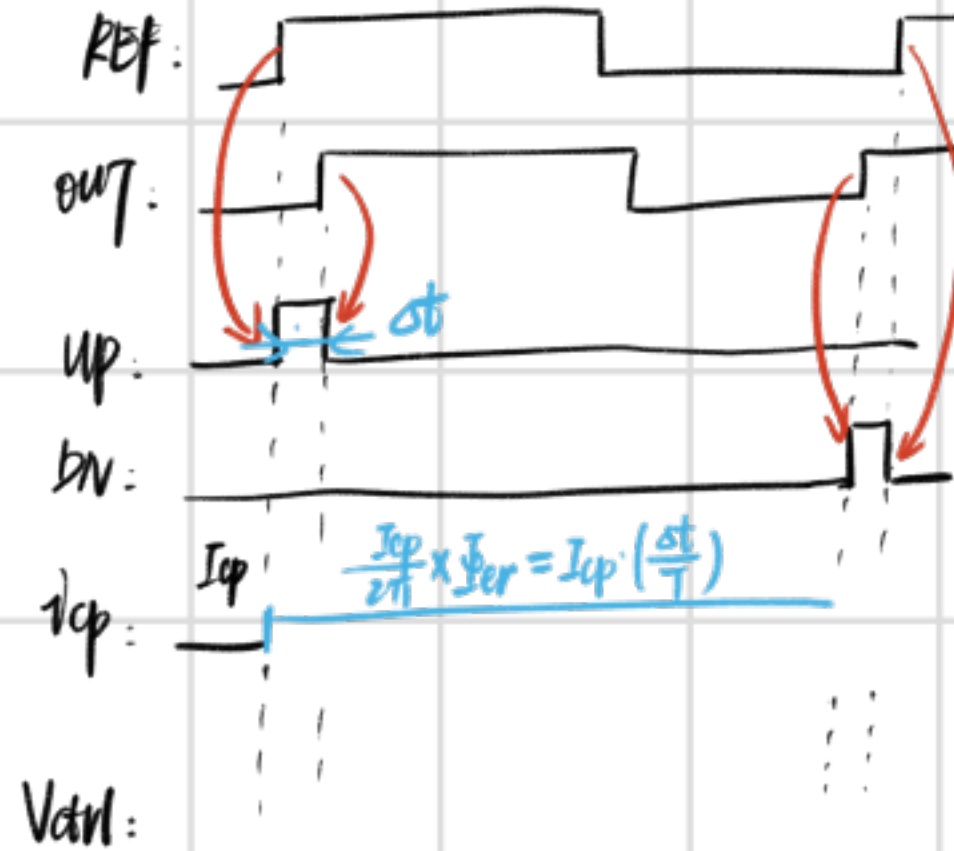
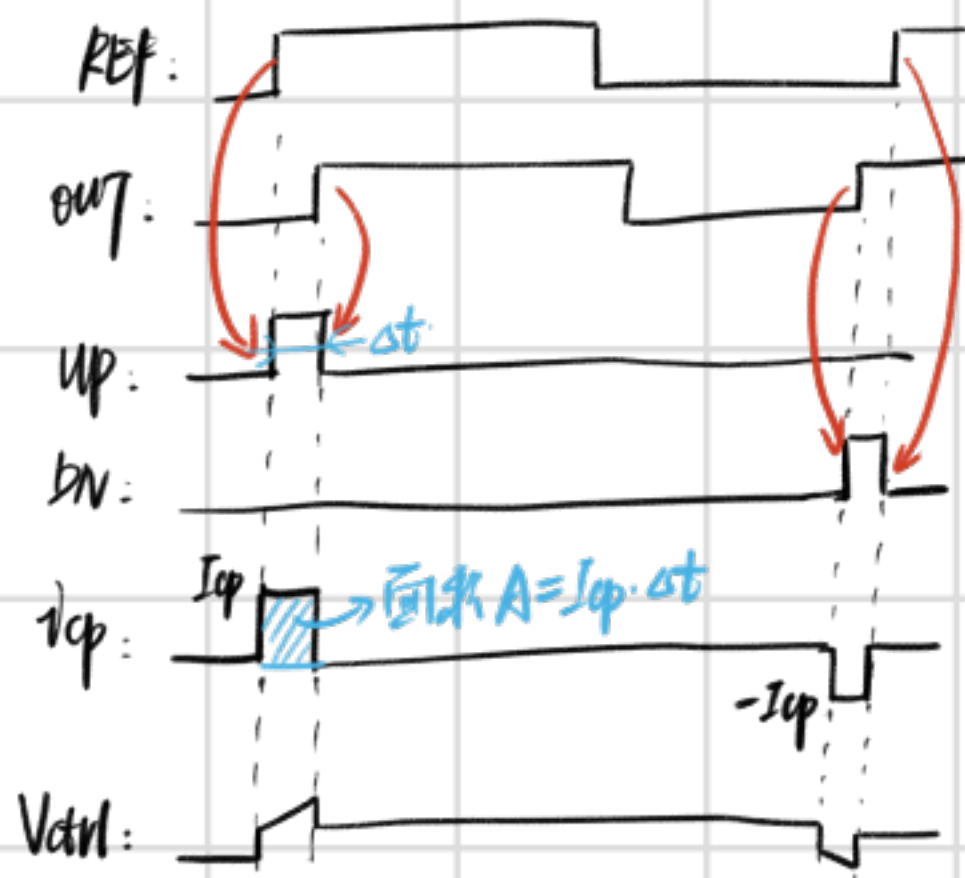
视为 PFD 和 CP 的结合的等效增益的增益

Small signal and Stability Analysis of Type-II order 2 Charge Pump PLL



Small signal block diagram of CP-PLL

Combined gain of PFD & CP = $\frac{I_{cp}}{2\pi}$

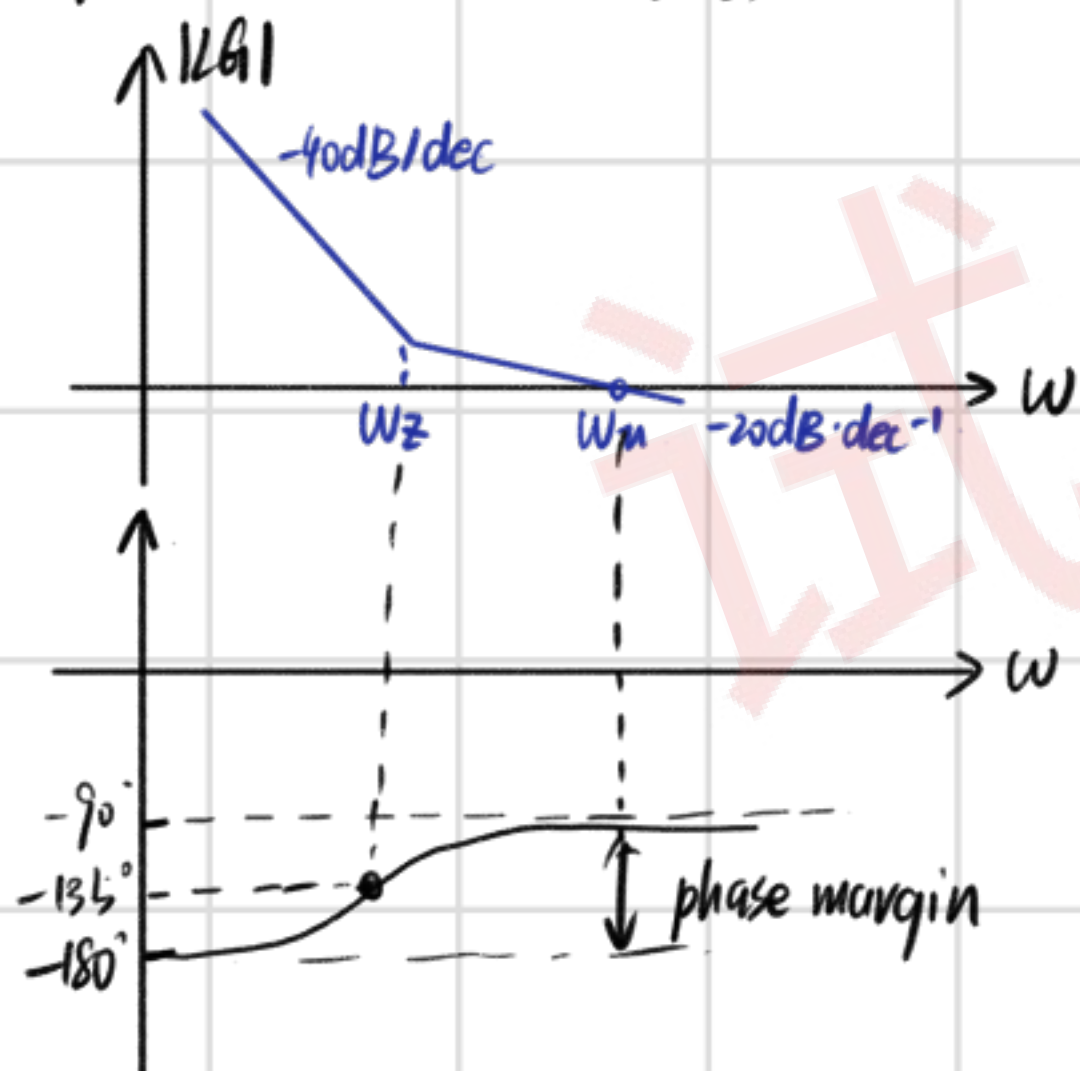


Small-signal Analysis of CP-PLL:

$$LG(s) = \frac{I_{cp}}{2\pi} \left(R + \frac{1}{sC_1} \right) \cdot \frac{K_{vco}}{s} = \frac{I_{cp} K_{vco}}{2\pi s^2 C_1} (1 + sRC_1)$$

$$\omega_{p1} = \omega_{p2} = 0, \omega_z = \frac{1}{RC_1}$$

$$\angle LG(j\omega) = -180^\circ + \arctan\left(\frac{\omega}{\omega_z}\right) = -180^\circ + \arctan(\omega RC_1)$$



phase margin: $\Phi_m = -180^\circ - (\angle LG(j\omega_u))$, Unity gain frequency ω_u , $|LG(j\omega_u)| = 1$
 $= \arctan(\omega_u RC_1)$

$$|LG(j\omega_u)| = 1$$

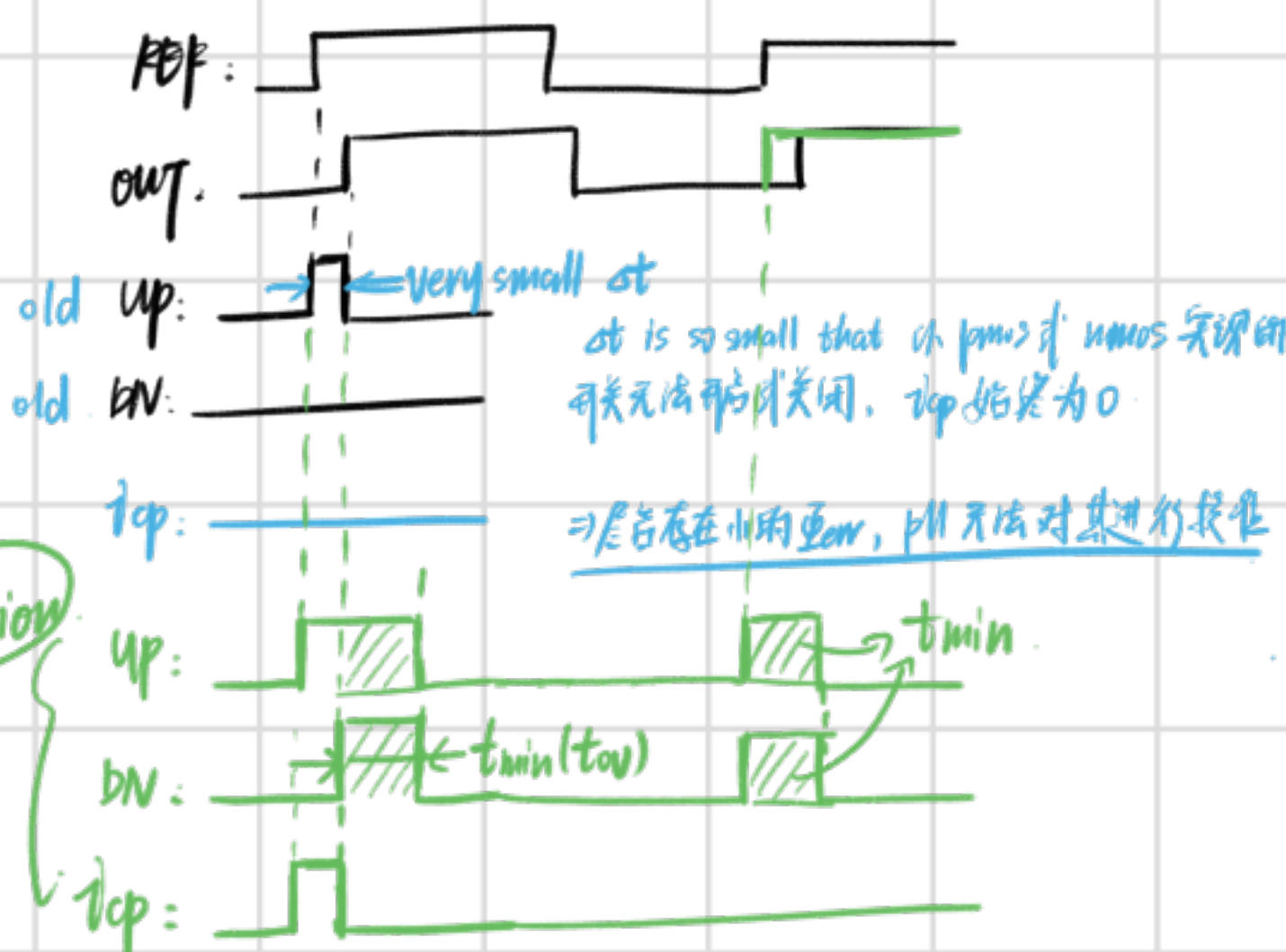
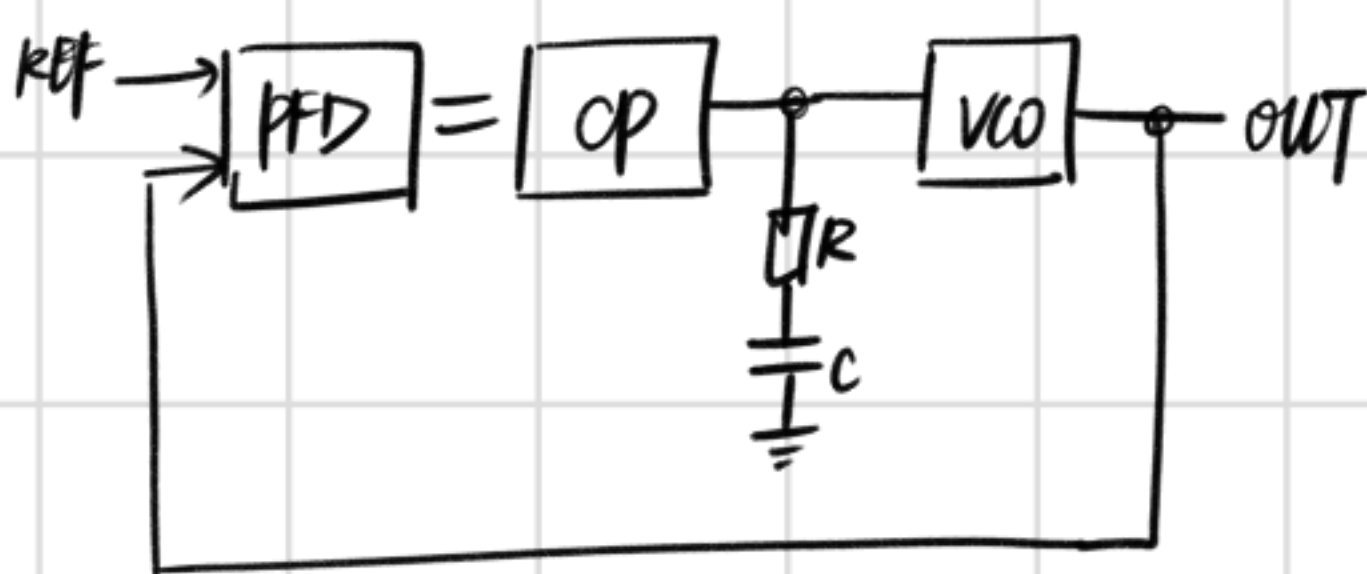
$$\text{Approximation: } |LG| = 1, \left| H \frac{j\omega_u}{\omega_z} \right| \approx \left| \frac{\omega_u}{\omega_z} \right| \gg 1$$

$$\frac{I_{cp} \cdot K_{vco}}{2\pi \omega_u^2 C_1} \times \omega_u RC_1 = 1$$

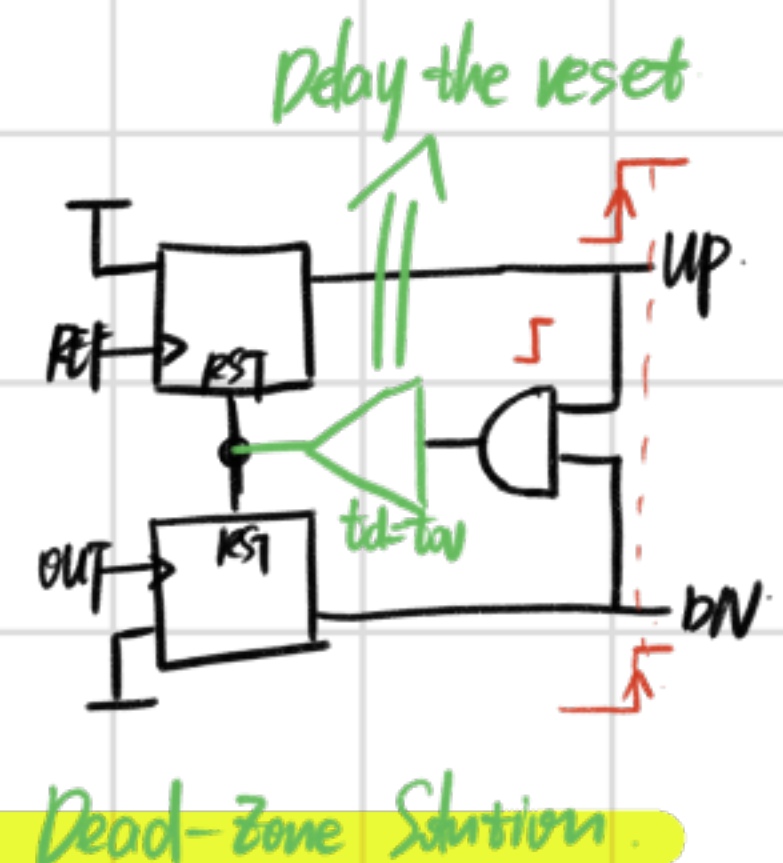
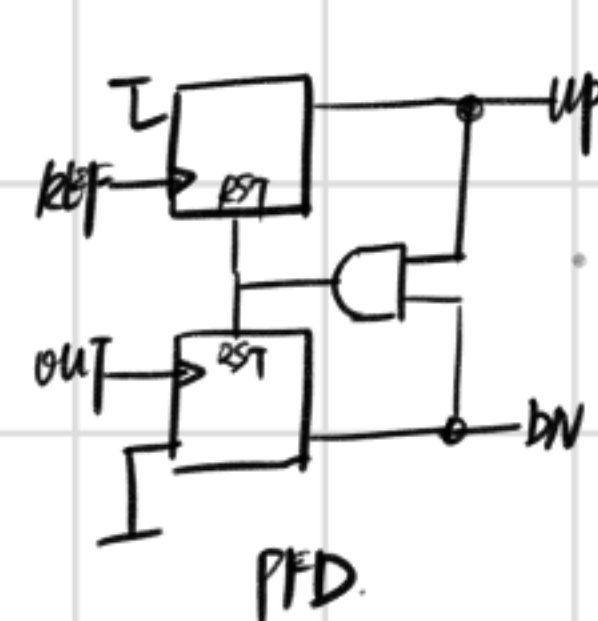
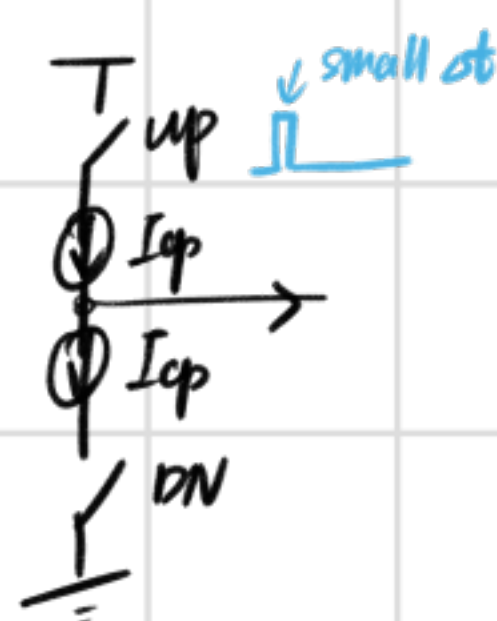
$$\omega_u = \frac{I_{cp} \cdot K_{vco} R}{2\pi}$$

$$\Phi_m = \arctan\left(\frac{I_{cp} \cdot K_{vco} R}{2\pi} \cdot RC_1\right), \text{ stable !!}$$

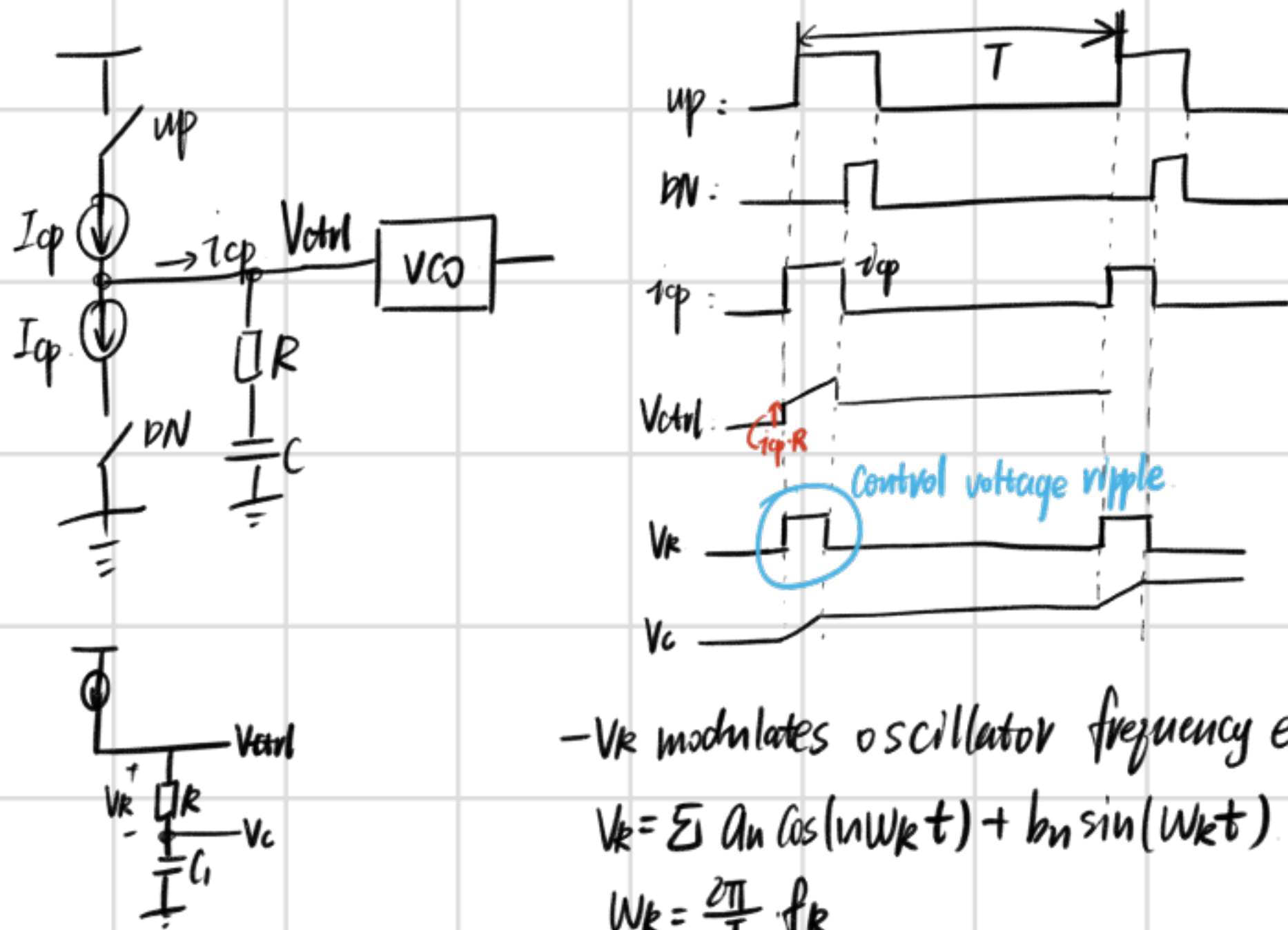
Problems in Charge-Pump PLL - Dead Zone in PFD



solution



Problems in Charge-Pump PLL — Reference Spur

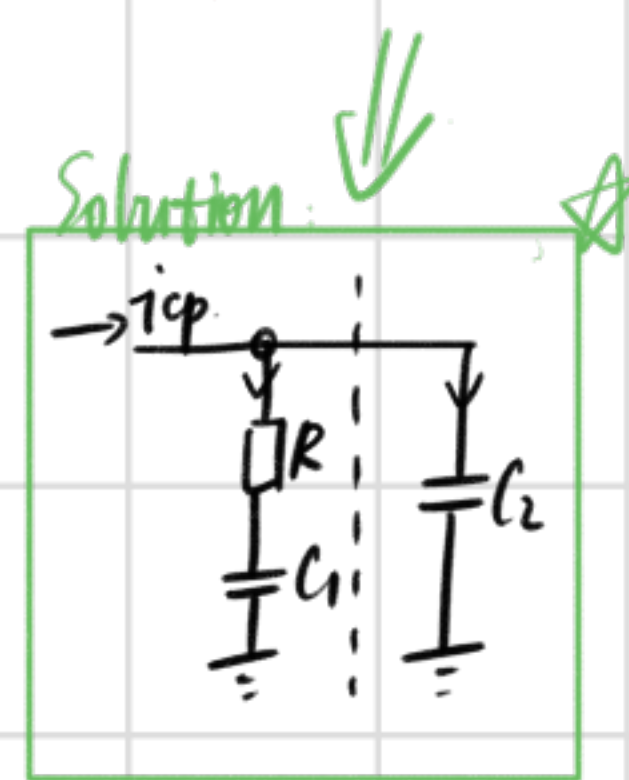


— V_k modulates oscillator frequency every T_k .

$$V_k = \sum a_n \cos(n\omega_k t) + b_n \sin(n\omega_k t)$$

$$\omega_k = \frac{2\pi}{T_k} f_k$$

— $(\omega_{out} \pm \omega_k)$ Reference Spur



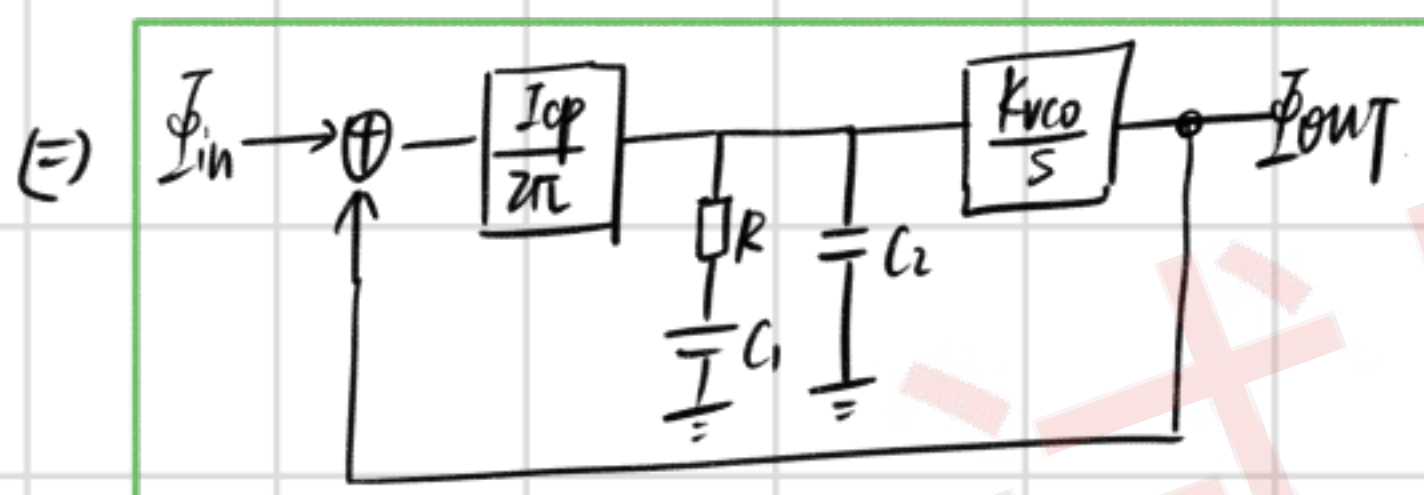
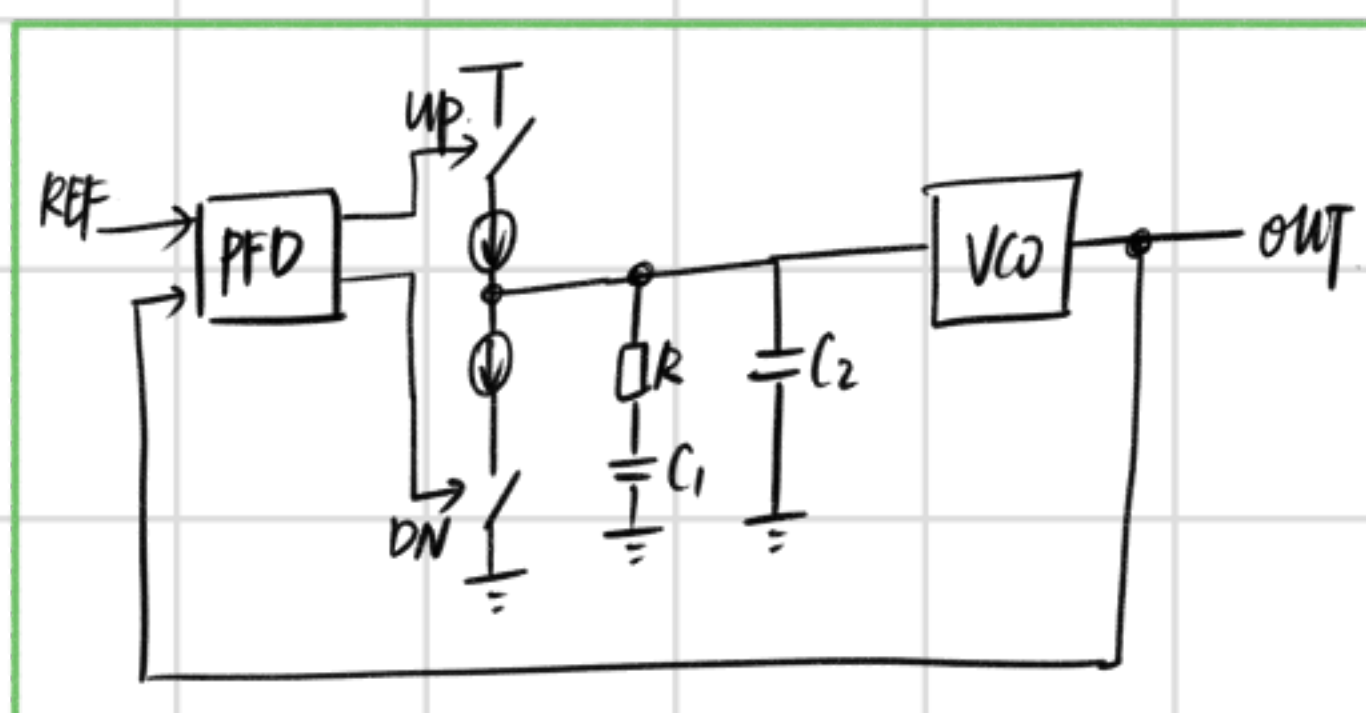
$$|j\omega_k C_2| \ll |R + \frac{1}{j\omega_k C_1}|$$

$$\Rightarrow C_2 \gg \frac{C_1}{1 + j\omega_k C_1 R}$$

to reduce ripples in V_{ctrl}

C_2 is called ripple bypass capacitor.

引入 C_2 后, VCO 的控制电压
的纹波减小了,
但是频率特性变差



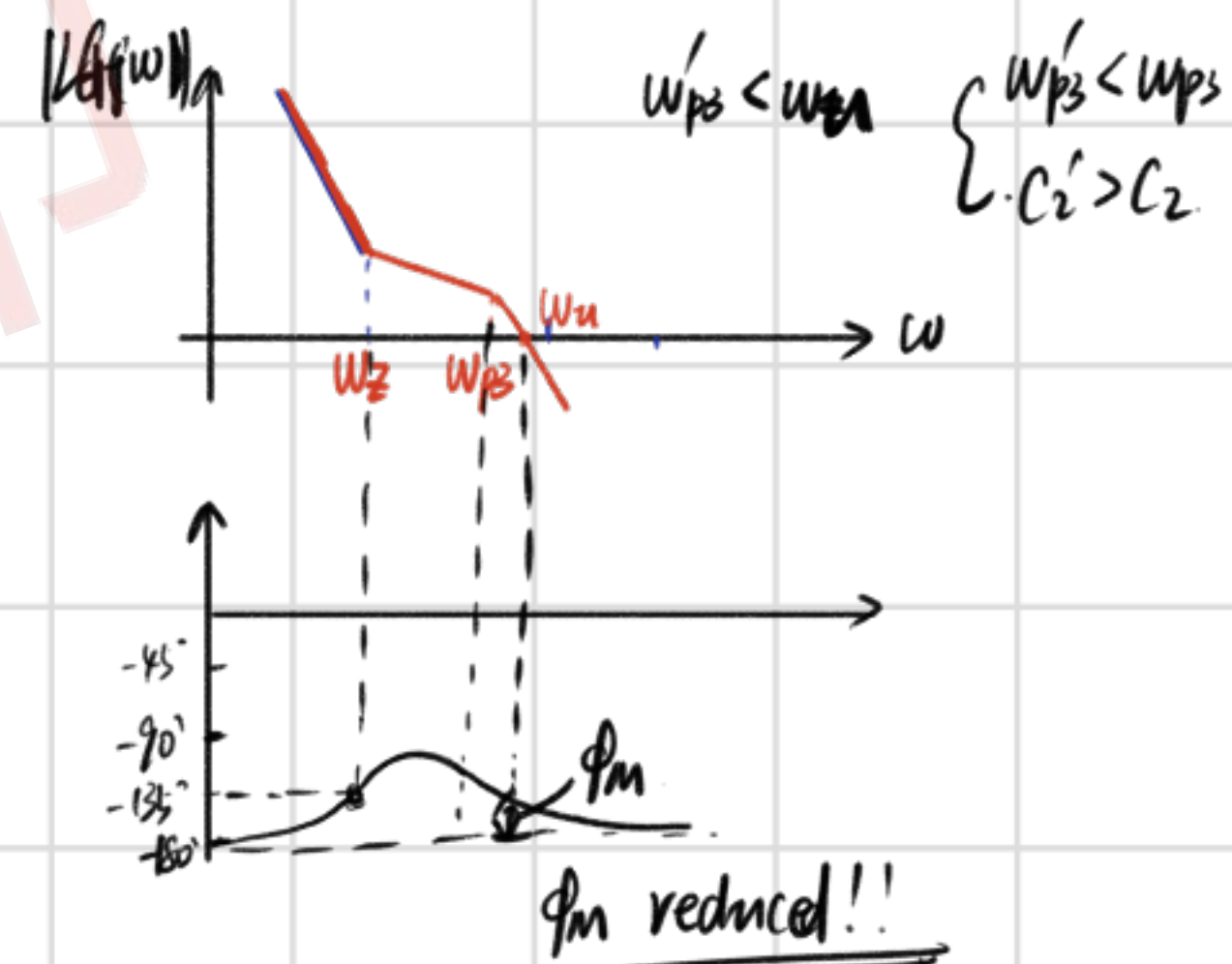
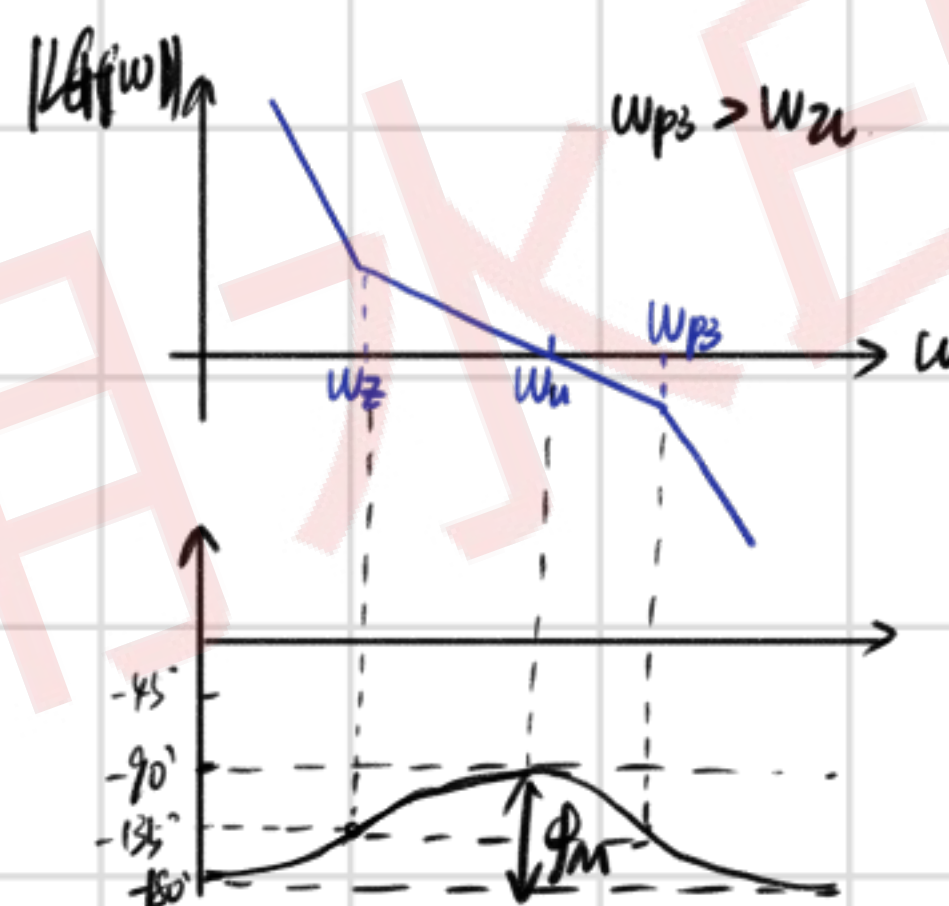
$$LG(s) = \frac{I_{cp}}{2\pi} \cdot \frac{H s R C_1}{1 + \frac{s R C_1 C_2}{C_1 + C_2}} \times \frac{K_{vco}}{s^2 (1 + z)}$$

$$\omega_{p1} = \omega_{p2} = 0, \quad \omega_{p3} = \frac{1}{RC_1 C_2} \quad \text{poles: } s_1 = s_2 = 0, s_3 = -\omega_{p3}$$

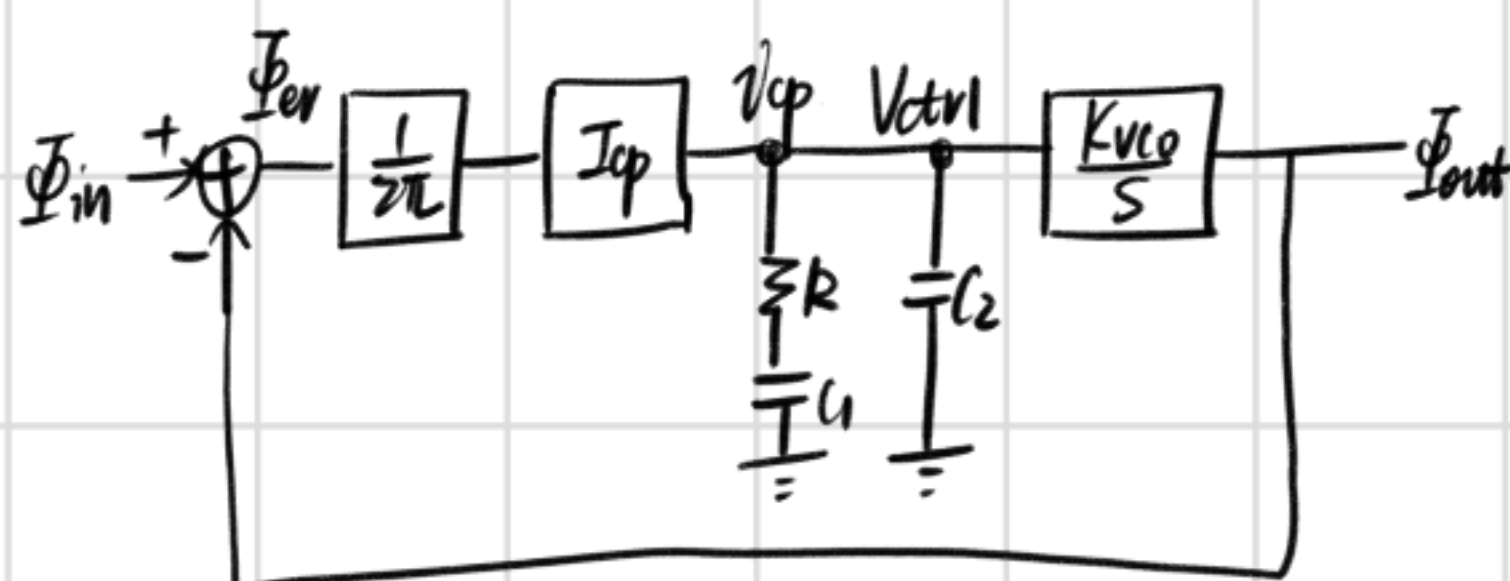
$$\omega_z = \frac{1}{RC_1} \quad \text{zero: } -\frac{1}{RC_1}$$

$$\Rightarrow LG(s) = \frac{I_{cp} K_{vco}}{2\pi s^2 (1 + z)} \cdot \frac{(1 + s/\omega_z)}{(1 + s/\omega_{p3})}$$

$$\angle LG(j\omega) = -180^\circ + \tan^{-1}\left(\frac{\omega}{\omega_z}\right) - \tan^{-1}\left(\frac{\omega}{\omega_{p3}}\right)$$



Design Procedure for Type-II Order 3 Charge Pump PLL



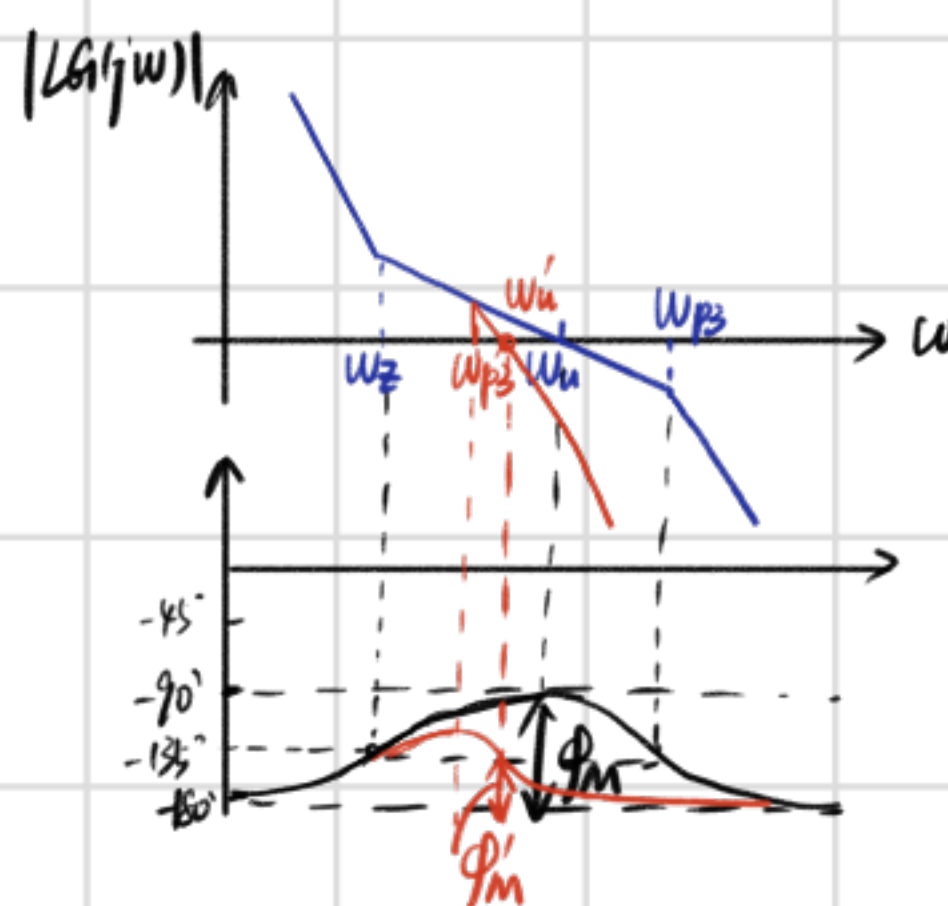
$$LG(s) = \frac{I_{cp}}{2\pi} \cdot \frac{H s R C_1}{1 + s \cdot \frac{R C_1 C_2}{C_1 + C_2}} \cdot \frac{K_{vco}}{s^2 (1 + z)}$$

$$\angle LG(j\omega) = -180^\circ + \arctan\left(\frac{\omega}{\omega_z}\right) - \arctan\left(\frac{\omega}{\omega_{p3}}\right)$$

$$\phi_m = \arctan\left(\frac{\omega}{\omega_z}\right) - \arctan\left(\frac{\omega}{\omega_{p3}}\right)$$

$$\omega_z = \frac{1}{RC_1}, \quad \omega_{p3} = \frac{1}{RC_1 C_2} = \omega_z \left(\frac{C_1}{C_2} + 1\right)$$

$$\phi_m = \tan^{-1}\left(\frac{\omega_u}{\omega_z}\right) - \tan^{-1}\left(\frac{\omega_u}{\omega_{p3}}\right)$$



the change in the phase margin at the unity frequency
is much less than when change to the unity
frequency by some amount.

$$|LG(j\omega_u)| = 1$$

Assuming, $\omega_z \ll \omega_u \ll \omega_{p3}$

$$\lim_{\omega \rightarrow \infty} \frac{d\Phi_m}{d\omega} = 0 \Rightarrow \frac{1}{H \left(\frac{\omega_n}{\omega_z} \right)^2} \cdot \frac{1}{\omega_z} - \frac{1}{1 + \left(\frac{\omega_n}{\omega_p} \right)^2} \cdot \frac{1}{\omega_p} = 0$$

$$\Rightarrow \omega_n^2 = \omega_z \omega_p$$

$$= \omega_z^2 \left(1 + \frac{C_1}{C_2} \right)$$

$$\Rightarrow \omega_n = \omega_z \sqrt{1 + \frac{C_1}{C_2}}$$

$$\Phi_m = \arctan \left(\sqrt{1 + \frac{C_1}{C_2}} \right) - \arctan \left(\frac{1}{\sqrt{1 + C_1/C_2}} \right)$$

$$K_c = \frac{C_1}{C_2} = 2 \left(\tan^2 \Phi_m + \tan \Phi_m \sqrt{1 + \tan^2 \Phi_m} \right)$$

Design Procedure:

1. Given ω_n and Φ_m

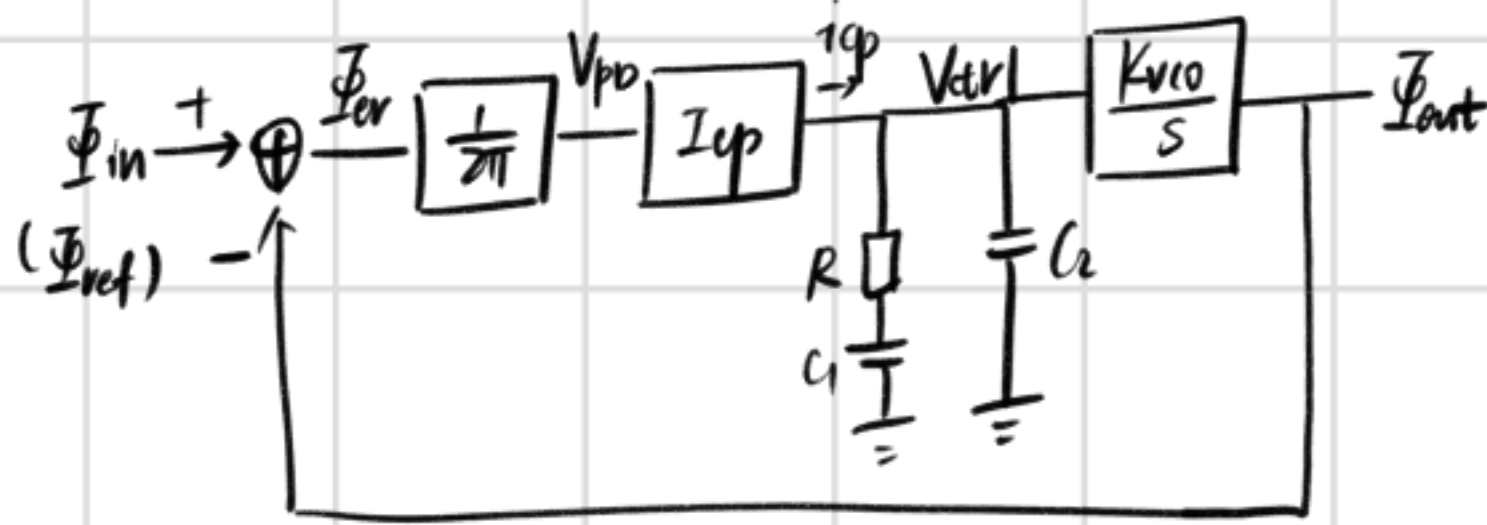
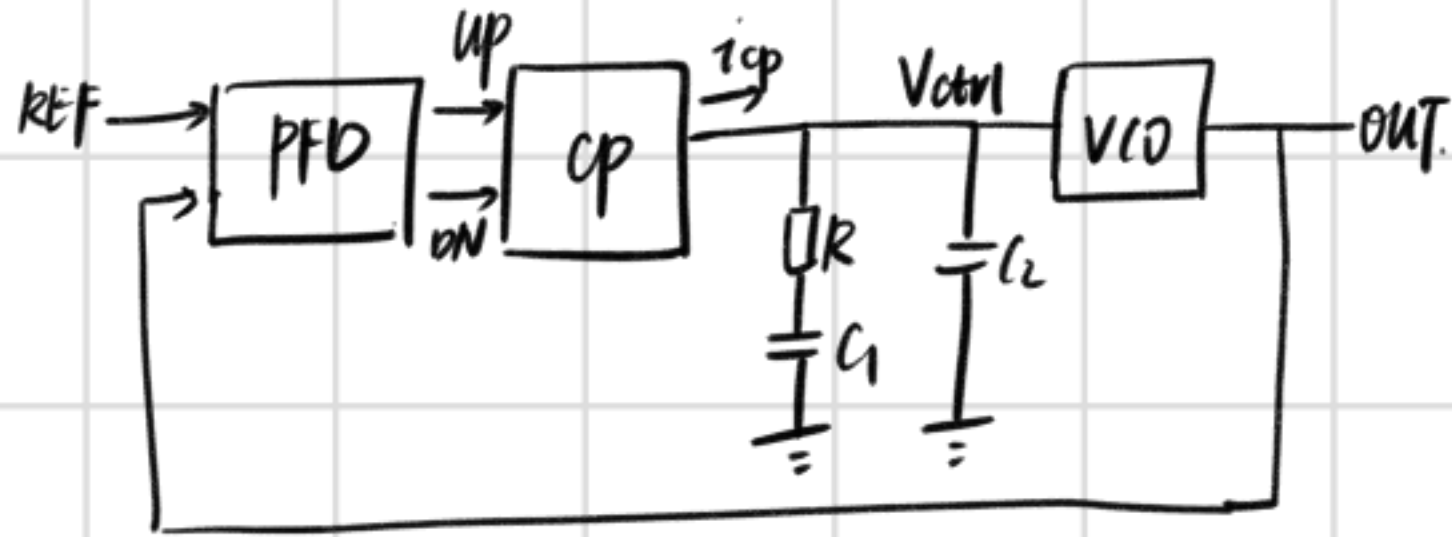
$$2. K_c = \frac{C_1}{C_2} = 2 \left(\tan^2 \Phi_m + \tan \Phi_m \sqrt{1 + \tan^2 \Phi_m} \right)$$

$$3. \omega_z = \frac{\omega_n}{\sqrt{1 + \frac{C_1}{C_2}}}$$

$$4. \text{Choose } R, C_1 = \frac{1}{\omega_z R}, C_2 = \frac{C_1}{K_c}$$

$$5. |G(s)| = 1 \Rightarrow I_{cp} = \frac{2\pi C_2}{K_{vco}} \omega_n^2 \sqrt{(\omega_p^2 + \omega_n^2) / (\omega_z^2 + \omega_n^2)}$$

Design Procedure for Charge Pump Clock Multiplier



Given Unity Gain Frequency ω_n , Phase Margin Φ_m

$$1. K_c = \frac{C_1}{C_2} = 2 \left(\tan^2 \Phi_m + \tan \Phi_m \sqrt{1 + \tan^2 \Phi_m} \right)$$

$$2. \omega_z = \frac{\omega_n}{\sqrt{1 + C_1/C_2}}$$

$$3. \text{Choose } R, C_1 = \frac{1}{\omega_z R}, C_2 = \frac{C_1}{K_c}$$

$$4. |G(j\omega_n)| = 1, \text{ find } K_{vco}$$

$$I_{cp} = \frac{2\pi(C_2 + C_1)}{K_{vco}} \omega_n^2 \sqrt{\frac{1 + \omega_n^2/\omega_p^2}{1 + \omega_n^2/\omega_z^2}}$$

$$\text{where } \begin{cases} \omega_p = \frac{1}{RC_2/(CH_2)} \\ \omega_z = \frac{1}{RC_1} \end{cases}$$

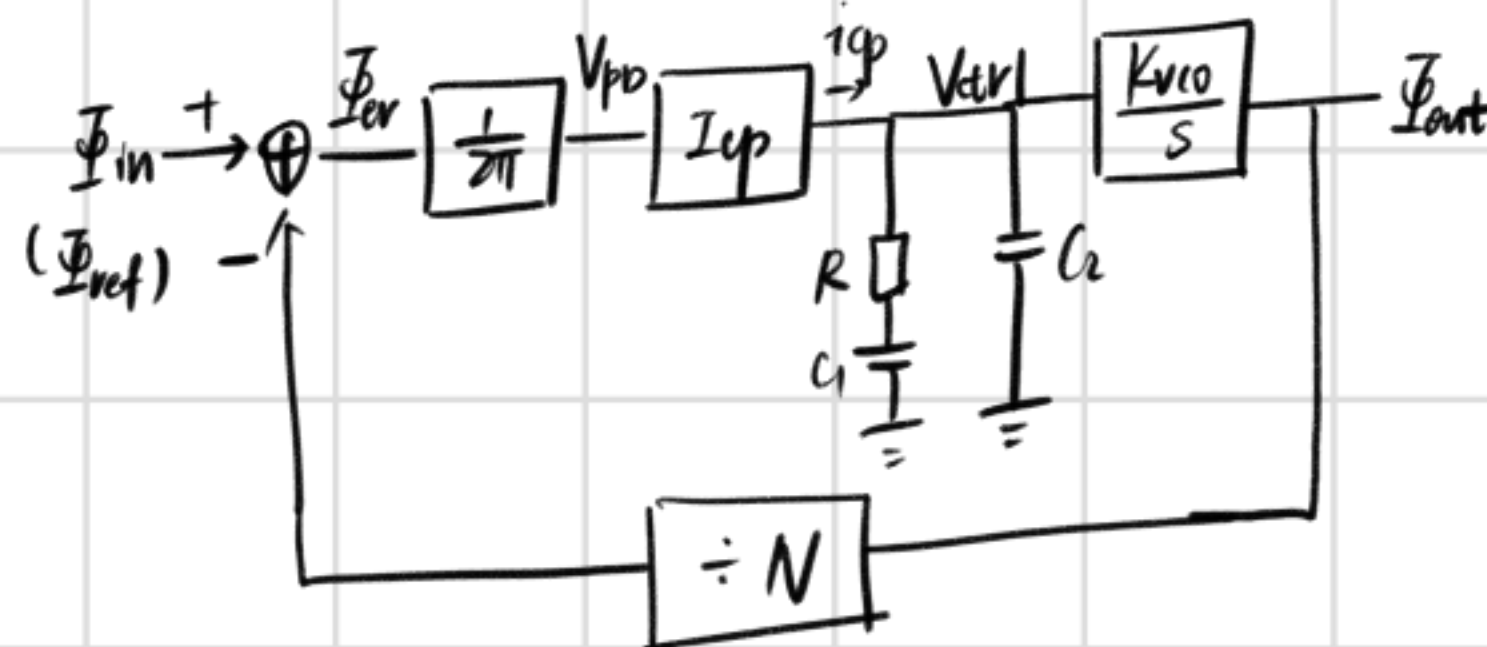
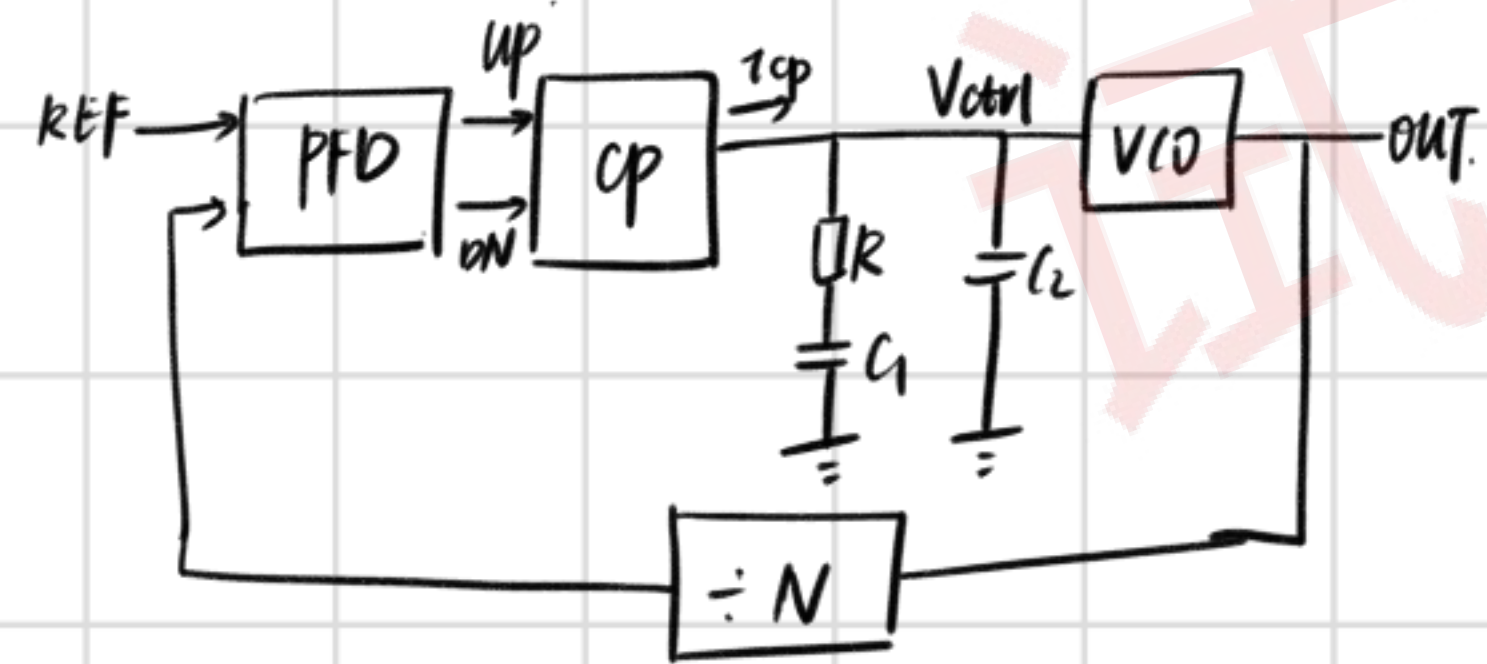
Given ω_n, Φ_m :

$$1. \text{find } \frac{C_1}{C_2}$$

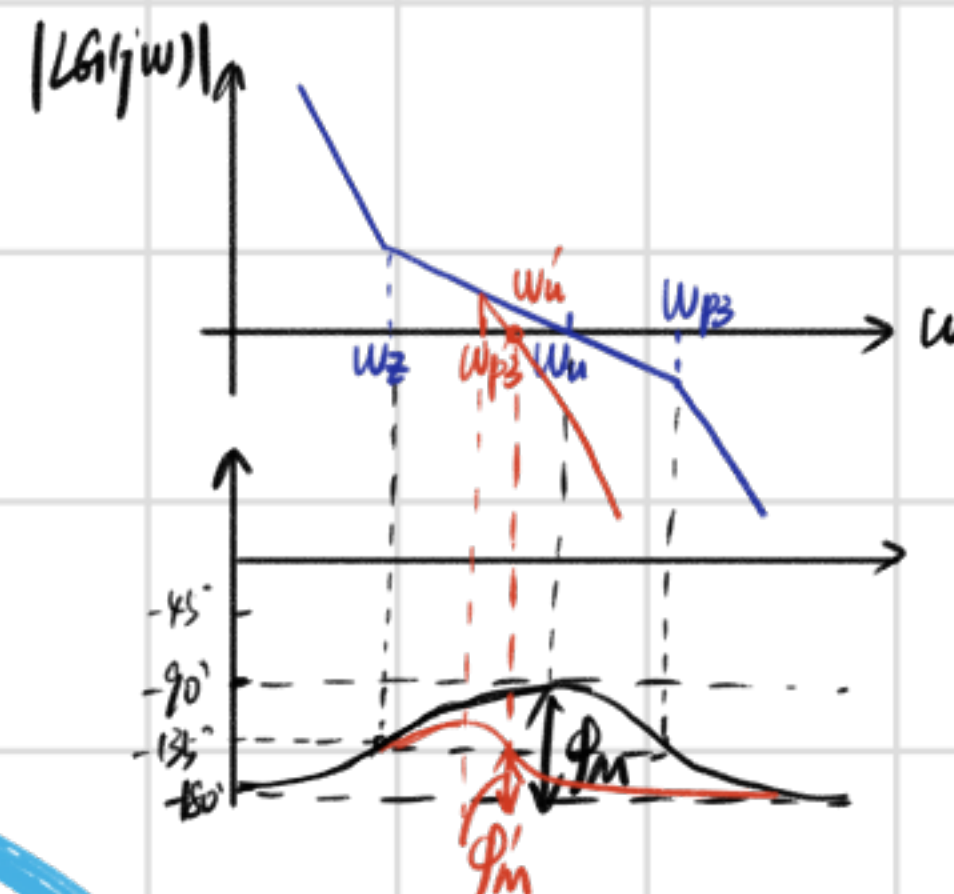
$$2. \text{find } \omega_z$$

$$3. \text{Choose } R, \text{ get } C_1, C_2$$

$$4. \text{Given } K_{vco}, I_{cp} = N \cdot \frac{2\pi(C_1 + C_2)}{K_{vco}} \omega_n^2 \times \sqrt{\frac{1 + \omega_n^2/\omega_p^2}{1 + \omega_n^2/\omega_z^2}}$$



$$|G(s)| = \frac{I_{cp}}{2\pi} \cdot \frac{1}{s(CH_2)} \cdot \frac{HSRC_1}{HSRC_1C_2/(CH_2)} \cdot \frac{K_{vco}}{s} \cdot \frac{1}{N}$$



the change in the phase margin at the unity frequency is much less than when change to the unity frequency by some amount.

$$\frac{I_{cp}}{2\pi} \cdot \frac{1}{\omega_n^2(CH_2)} \cdot \frac{\omega_n/\omega_z}{1} \cdot \frac{K_{vco}}{N} \approx 1$$

$$\omega_n = \frac{I_{cp}}{2\pi} \cdot \frac{RC_1}{CH_2} \cdot \frac{K_{vco}}{N} = \frac{I_{cp} \cdot R \cdot K_{vco}}{2\pi N} \cdot \frac{1}{1 + \frac{C_2}{C_1}}$$

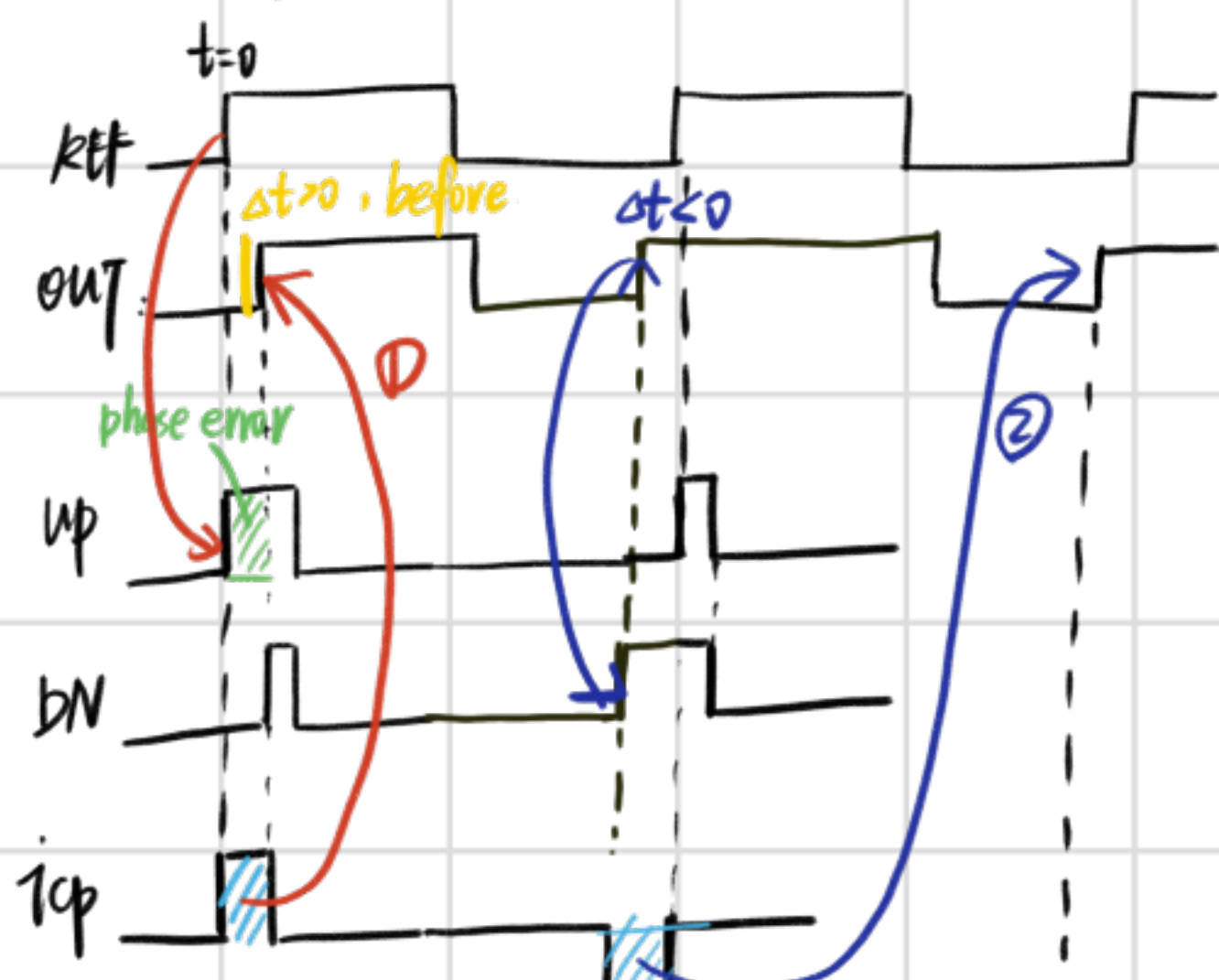
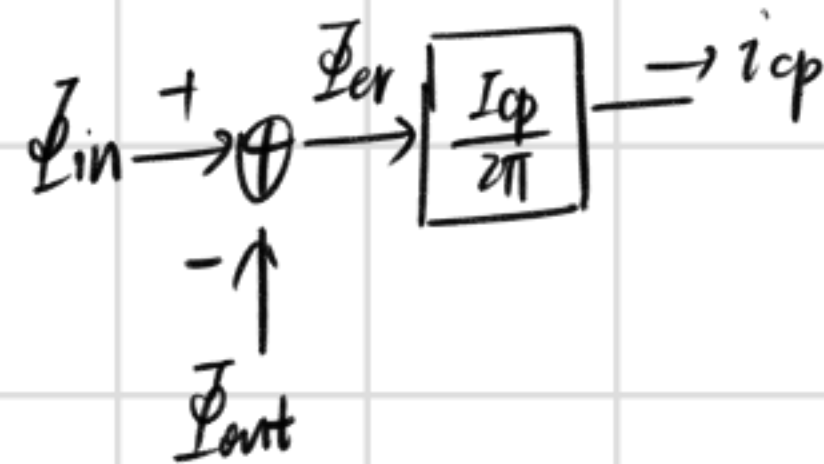
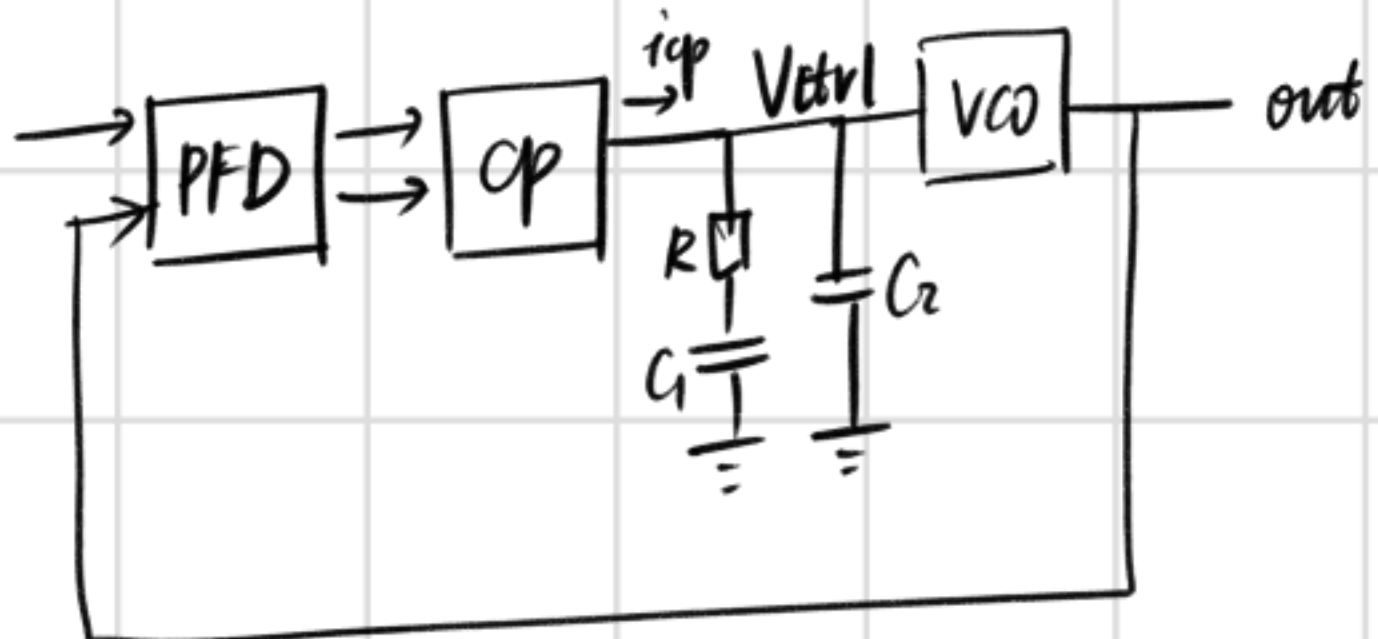
$$\omega_n = \frac{I_{cp} \cdot R \cdot K_{vco}}{2\pi N \left(1 + \frac{C_2}{C_1} \right)}$$

$$|G(j\omega_n)| = 1$$

Assuming, $\omega_z \ll \omega_n \ll \omega_p$

Sources of non-linearities in CP-PLL

1.

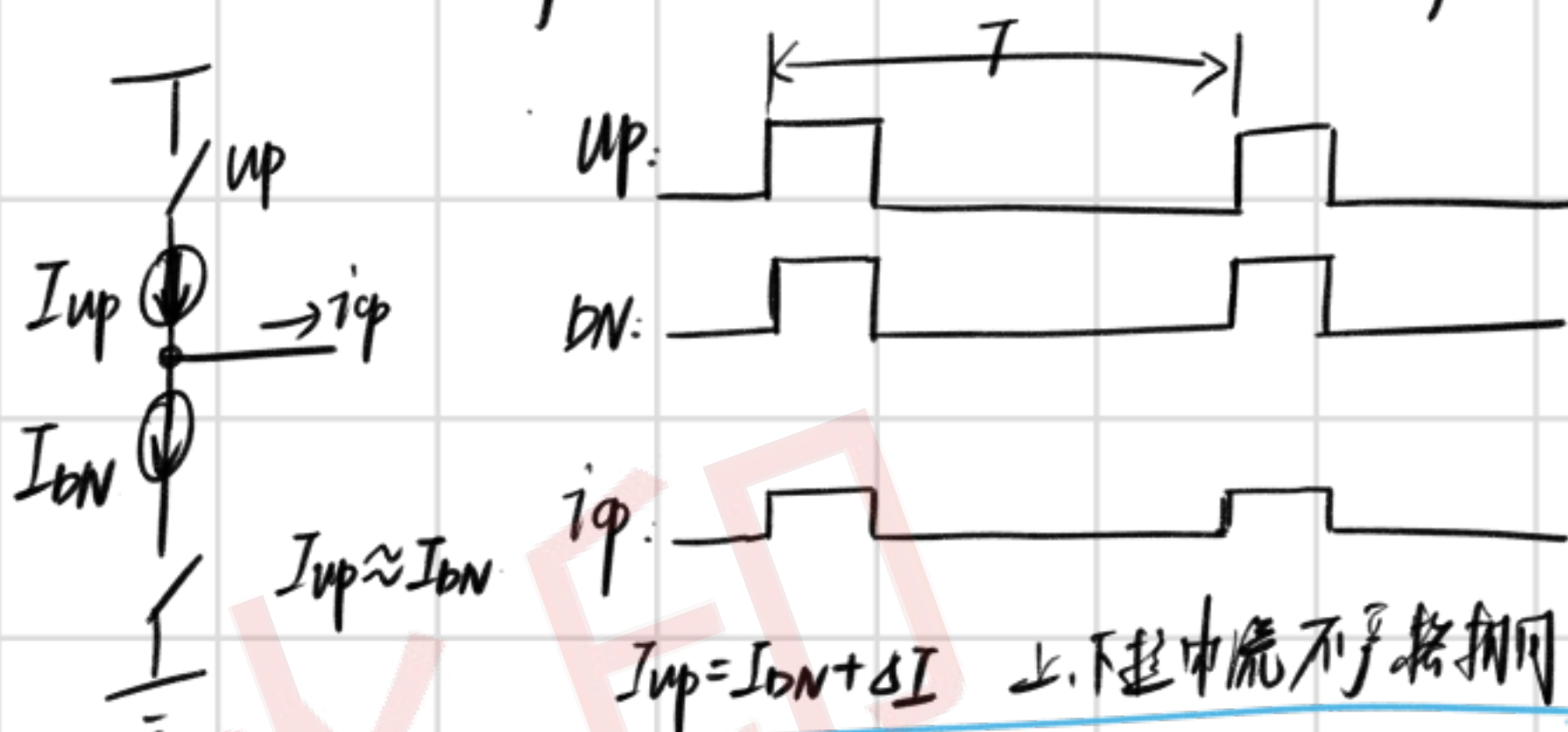


- if at time instant $t=0$, phase error between input and output is $2\pi \cdot \frac{\Delta t}{T}$, the rising edge of output will come before time Δt ($\Delta t > 0$).

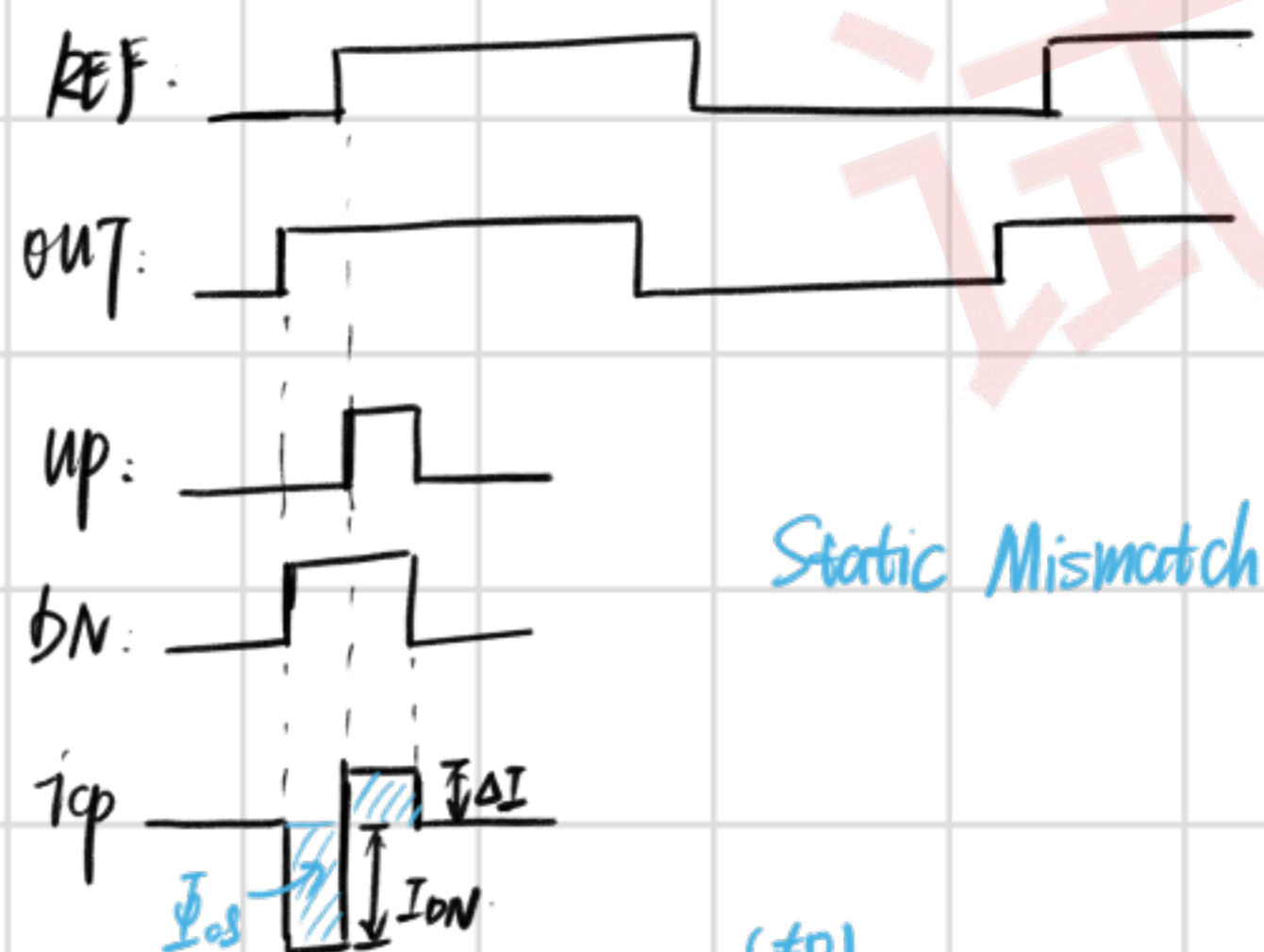
- Response of the PLL is different for the $+\Phi_e$ and $-\Phi_e$, with the same magnitude.

2. Dead ZONE in PFD

3. Static and dynamic mismatch in Charge-Pump



Assuming $I_{up} > I_{dn}$

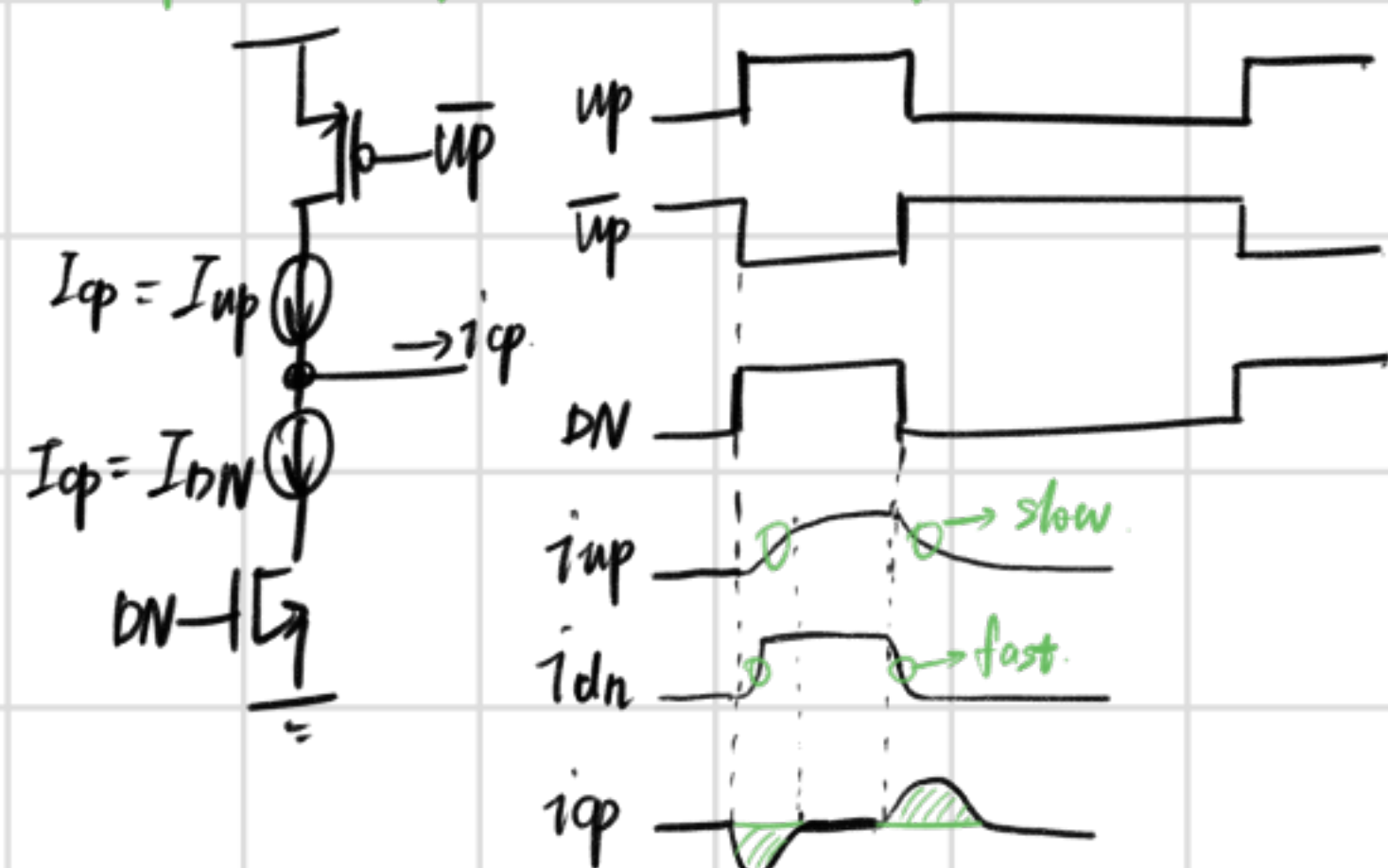


Static Mismatch

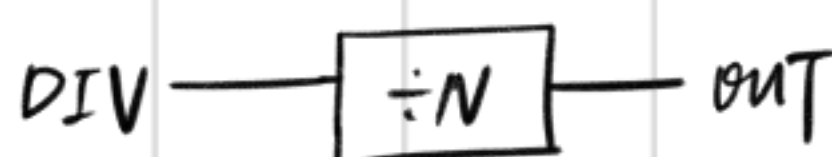
the PLL with locked in a Φ_{offset} , 此时上下区面积相等

$$-(I_{up} - I_{dn}) = f(V_{ctrl})$$

Response from pmos switches is different from the nmos switches



4. Delay in the loop



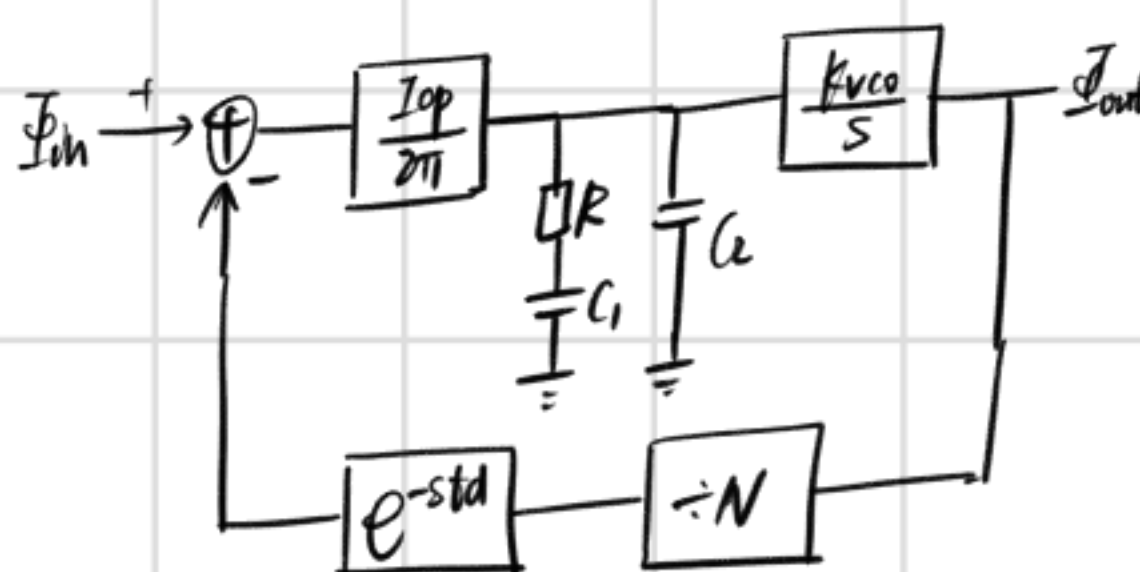
eg: $\div 2$



$$\Phi_{div}(t) = \frac{\Phi_{out}(t)}{2} \times$$

$$\Rightarrow \Phi_{div}(t) = \frac{1}{2} \Phi_{out}(t - t_d)$$

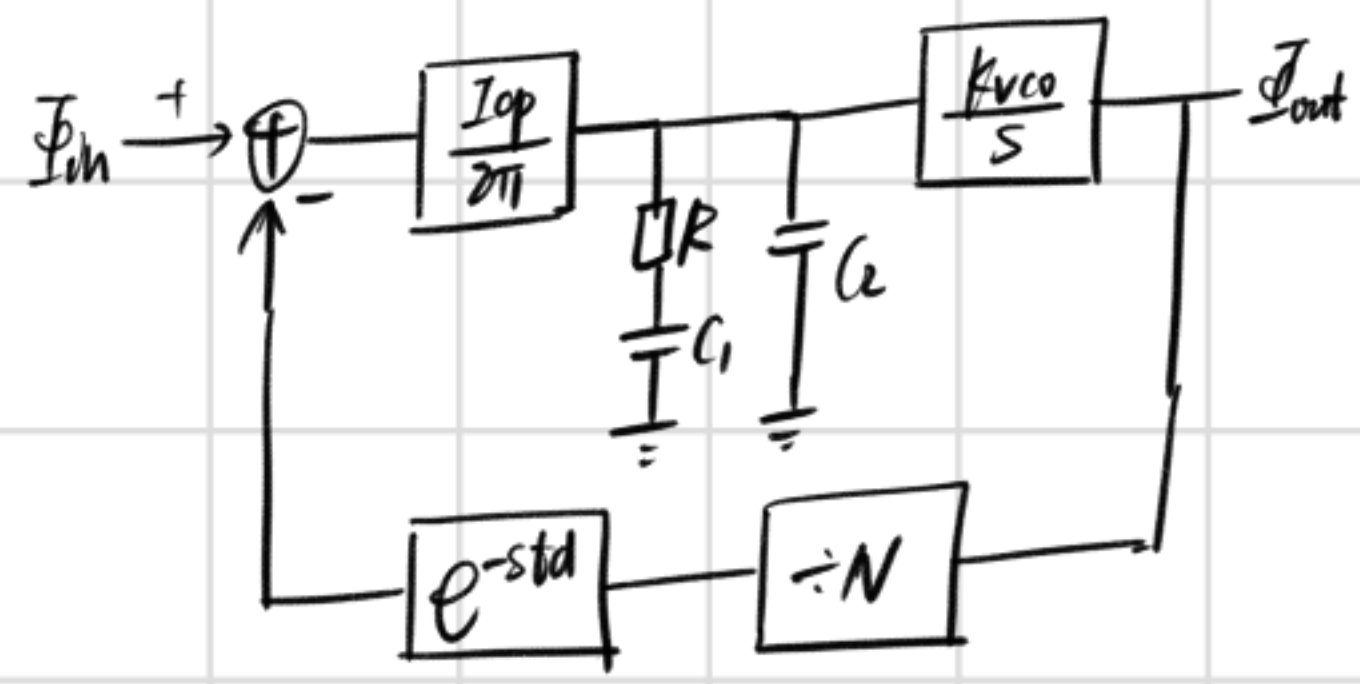
$$\Rightarrow \Phi_{div}(s) = \frac{1}{N} \Phi_{out}(s) e^{-s t_d}$$



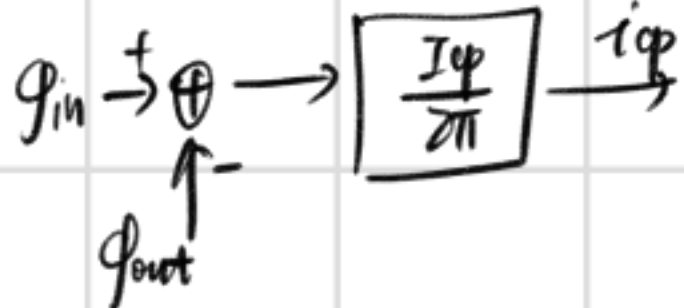
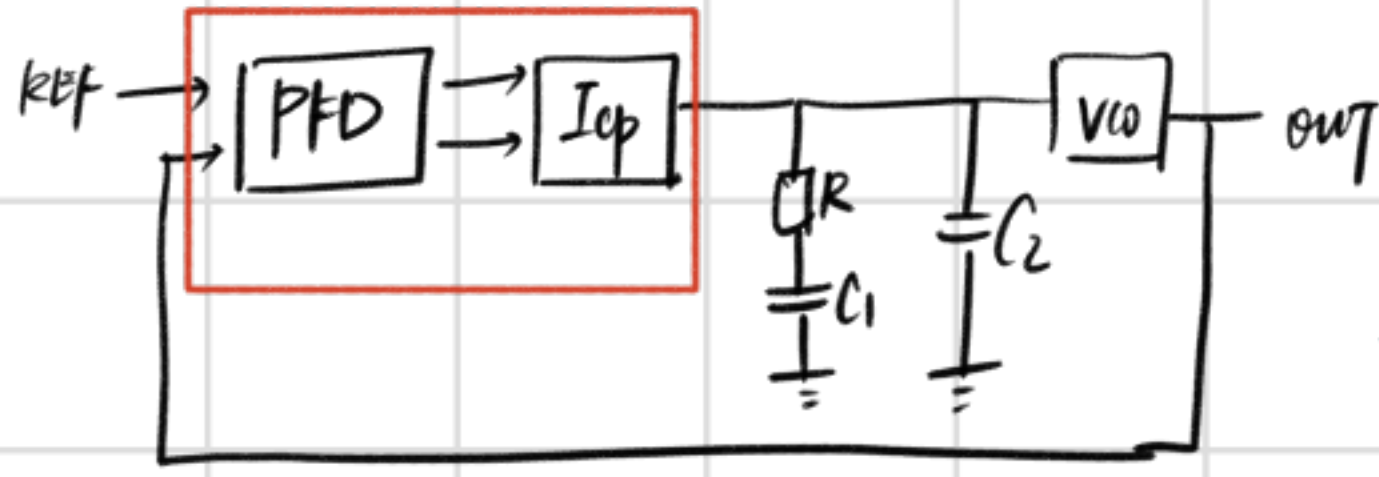
$$LG(s) = \frac{I_{cp}}{2\pi} \frac{K_{vco}}{s^2(C_1+C_2)N} \cdot \left(\frac{Hs/W_z}{Hs/W_p3} \right) \cdot e^{-std}$$

$$\angle LG(s) = -180^\circ + \tan^{-1}\left(-\frac{W}{W_z}\right) - \tan^{-1}\left(\frac{W}{W_p3}\right) - \tan^{-1}(Wtd)$$

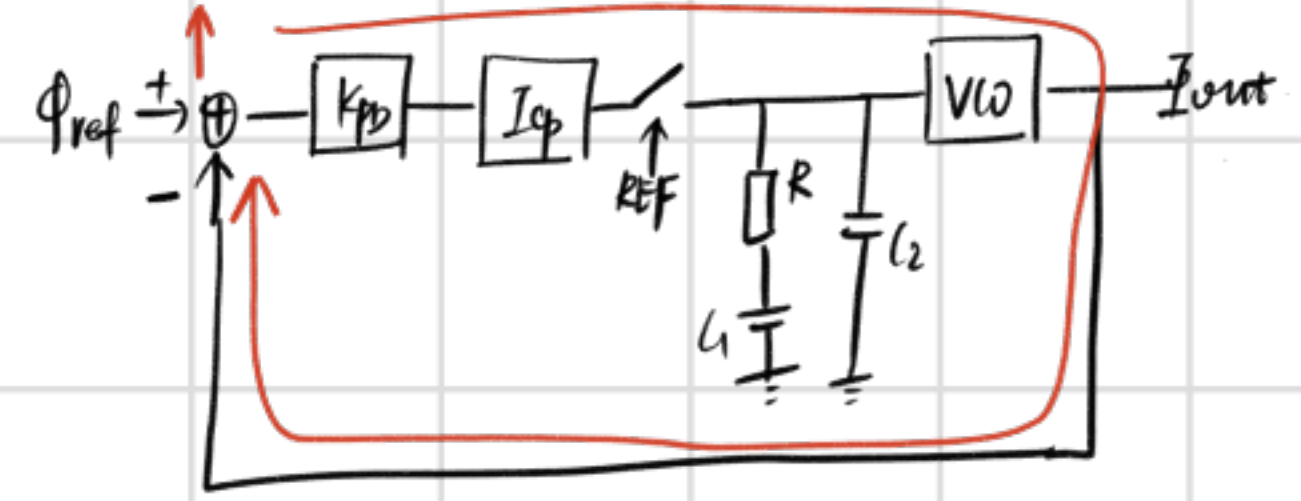
$$\Phi_M = \tan^{-1}\left(\frac{W_n}{W_z}\right) - \tan^{-1}\left(\frac{W}{W_p3}\right) - \tan^{-1}(Wtd) \quad \text{相位裕度恶化}$$



Effect of Sampling:

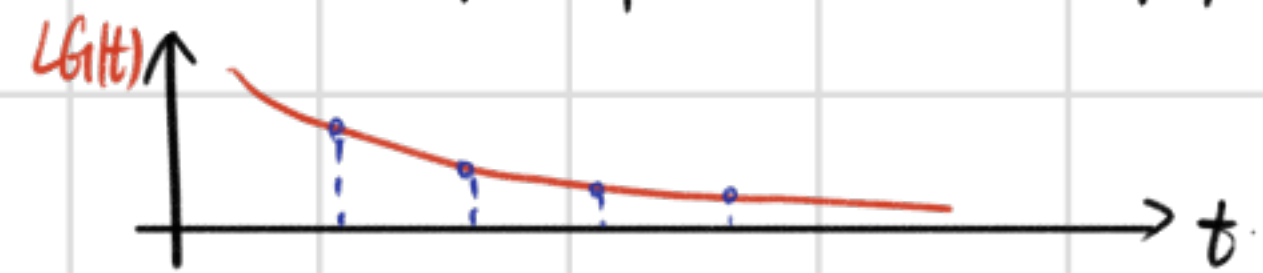


- PFD operates only on rising edge of clock
- No feedback only get the rising edge of ϕ_{ref} and rising edge of V_{co} between two positive edges.
- One reference clock delay



- Impulse Invariance Method:

- Time-domain samples of loop gain $LG(t)$
- Pick samples for $LG(t)$ at $t = n \cdot T_{ref}$



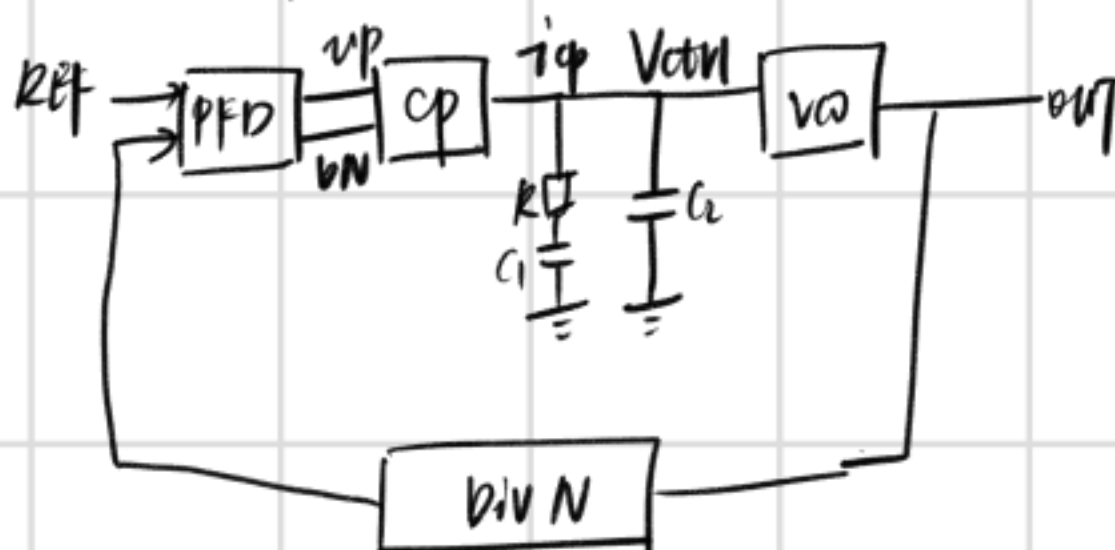
- Discrete loop gain samples at $t = n \cdot T_{ref}$

$$LG(z) = \frac{k \left[z^2 \left(\frac{C_1(1-a)}{C_1+C_2} + \frac{2\pi}{RC_1W_{ref}} \right) - z \left(\frac{C_1(1-a)}{C_1+C_2} + \frac{2\pi}{RC_1W_{ref}} a \right) \right]}{z^3 - z^2(2+a) + z(1+2a) - a} \quad \left(\approx \frac{I_{cp}}{2\pi} \frac{K_{vco}}{s^2(C_1+C_2)N} \cdot \left(\frac{Hs/W_z}{Hs/W_p3} \right) \right)$$

here, $a = \exp(-W_p3 T_{ref})$, $k = \frac{K_{vco} I_{cp} R C_1}{W_{ref} (C_1+C_2)}$

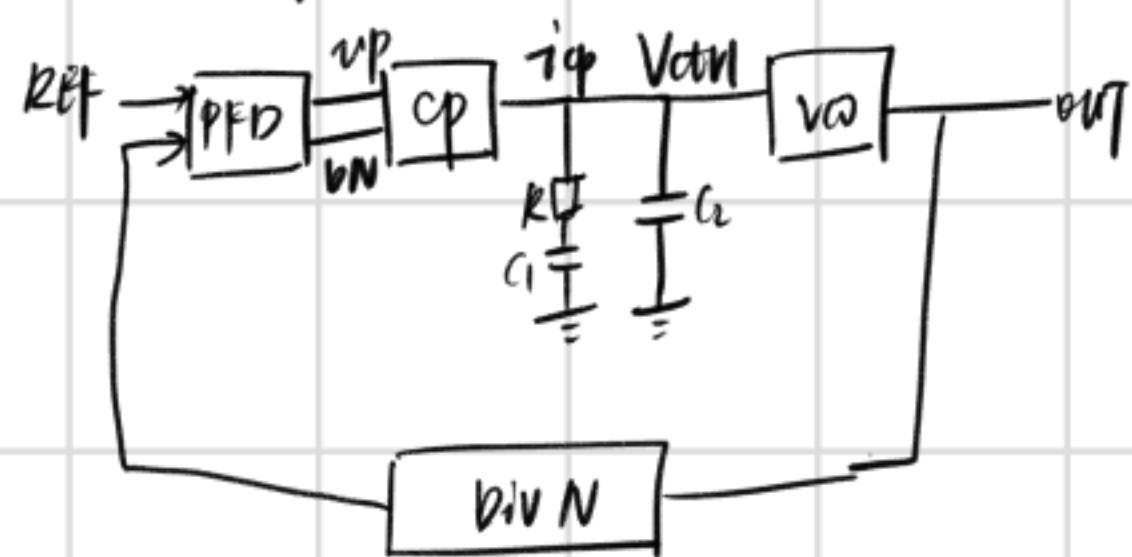
- if $\frac{W_{ref}}{W_n} \gg 10$, then laplace-domain analysis holds to a good approximation
- as $\left(\frac{W_{ref}}{W_n}\right) \downarrow$, and approaches values ≤ 3.5 , the system starts to becoming unstable
- if $W_n > \frac{W_{ref}}{10}$, requires z-domain analysis.
- if $W_n \gg \frac{W_{ref}}{10}$, oppll remains unstable.

Noise Analysis in CP-PLL

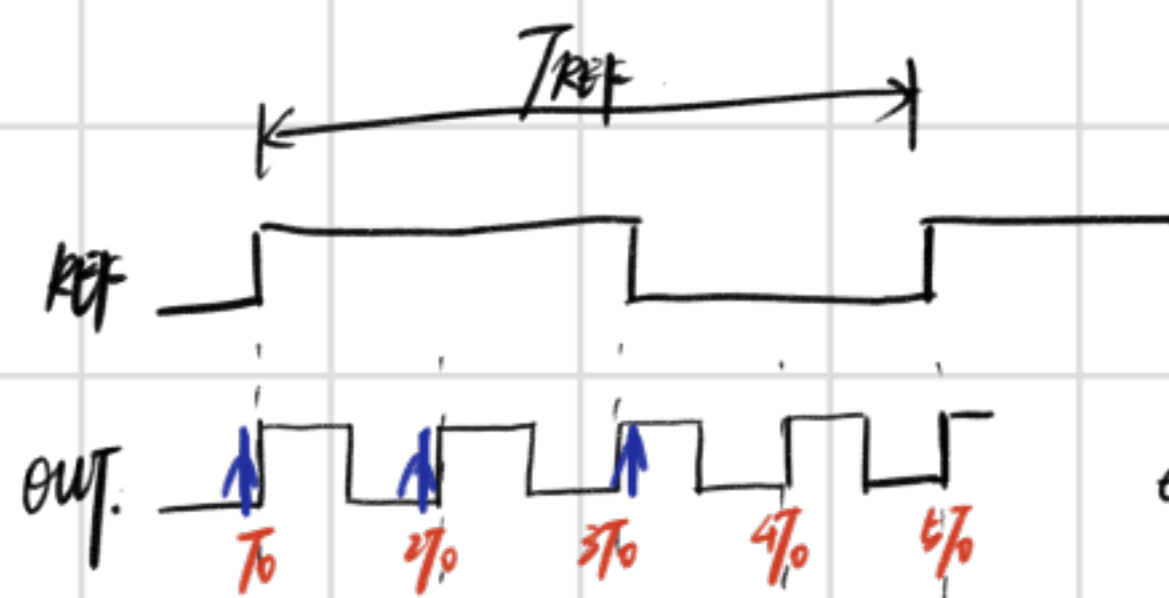


- R, C_1, C_2, I_{cp} for given W_n & Φ_M

Noise Analysis in CP-PLL



- R, C_1, C_2, I_q for given ω_n & $\mathcal{I}_m$
- Final output noise is a function of ω_n , for given loop components.
- $\mathcal{I}_m$ decides linear setting of the loop for any disturbance in PLL



eg. $T_{out} = \frac{T_{ref}}{N}$

t_i : time instant for i -th rising edge

$$\Delta t_i = t_i - (i-1)T_0$$

$$\mu_{\Delta t} = \sum \Delta t_i$$

$$\sigma_{\Delta t}^2 = \mathcal{E}[(\Delta t_i - \mu)^2]$$

σ : RMS 抖动 (Jitter)

$\Delta t_{max} - \Delta t_{min}$: Peak-to-Peak jitter

实际上升和理想上升的差值

试用水印